Breaking Boundaries: A Novel Heuristic for Bundle Pricing with General Product inter-relatedness

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Abstract

This article introduces a groundbreaking heuristic algorithm for bundle pricing problems, a popular marketing strategy that offers discounted prices for product bundles. We propose the first general heuristic method to determine which product subsets to offer at what prices, with a proper performance across a wide range of conditions. The heuristic is designed to accommodate any level of product inter-relatedness and can start from any feasible point, providing a practical and adaptable framework. Our algorithm employs a pricing and assortment alternating approach to marginally reduce the gap with the optimal solution. We evaluate the effectiveness of our heuristics by comparing its performance analytically with well-known heuristics such as pure bundling and bundle size pricing. A thorough numerical study showcases the superior performance of the proposed heuristic compared to existing heuristics in the literature.

Keywords— cardinality bundling; bundle size pricing; component pricing; approximation algorithm; marketing strategy

1 Introduction

Bundle pricing is widely recognized as a powerful marketing strategy that contributes to revenue growth. The Harvard Business Review demonstrates its effectiveness by asserting that bundling and unbundling are the only ways to generate profits (Fox, 2014). Analytical and empirical studies have consistently provided evidence supporting the positive impact of bundling as a marketing strategy, demonstrating its potential to enhance profitability across diverse business (Olderog & Skiera, 2000).

A noteworthy gap in the literature regarding a general heuristic approach that can be widely applied persists despite the large body of research on bundle pricing. It underscores the importance of further research and the development of new methodologies. Hence, the present study aims to address this gap by introducing the first set of general heuristics that provide a systematic framework, enabling businesses to effectively enhance their profitability through bundle pricing.

Bundling research has predominantly focused on two product scenarios or asymptotically optimal algorithms: pure bundling and bundle size pricing. Pure bundling refers to offering a single bundle comprising all available products, while bundle size pricing involves determining bundle prices solely based on their size, without considering the specific elements included. However, these approaches may not be suitable for businesses with medium product ranges. Furthermore, existing asymptotic analyses have primarily concentrated on independent products, overlooking cases where products are correlated, complementary, or substitutable (Abdallah et al., 2021; Bakos & Brynjolfsson, 1999). Venkatesh and Kamakura (2003) demonstrated that the effectiveness of bundling strategies heavily relies on the interrelationships among products. Additionally, certain products serve general purposes, resulting in highly distinctive relationships (Shocker et al., 2004). For example, products like pagers and wireless telephones, as well as email and voicemail, exhibit varying levels of complementarity or substitutability. This complexity further complicates the bundling problem, underscoring the need for general heuristics.

Common business strategies may deviate from the concepts of pure bundling and bundle size pricing. To illustrate the real-world application of bundle pricing, let's consider examples from prominent companies. iTunes, for instance, allows users to choose between individual tracks or albums based on the characteristics of the products. Similarly, Amazon eBook, the world's largest online book seller, provides a range of bundles alongside individual titles, demonstrating a more flexible bundling strategy than pure bundling or bundle size pricing. For example, their subscription service Kindle Unlimited offers over a million titles for 9.99 per month, providing unlimited access to download and read books, although it may not include the latest releases or bestsellers (Price, 2021).

Major retailers like Wal-Mart and Amazon also adopt a flexible approach to bundling, based on the value of the products rather than relying solely on bundle size pricing or pure bundle. These companies offer various product combinations, taking into account consumer preferences and maximizing sales. For instance, during the back-to-school season, Wal-Mart may bundle school supplies such as backpacks, notebooks, and pens into a single package to attract budget-conscious parents and students. Grocery stores and supermarkets commonly offer bundle packages, such as two packs of cereal or two packs of frozen pizza, at a discounted price, showcasing bundling of substitutable products.

Despite the prevalence of bundle pricing across diverse industries, the restrictive assumptions and strategies offered by previous research limit their applicability in real-world business settings. This study aims to address these limitations by proposing a novel heuristic approach, the pricing-assortment heuristic (PAH). The PAH leverages the fundamental characteristics and principles of the bundle pricing problem, breaking it down into more manageable subproblems. Specifically, it iteratively addresses pricing and assortment optimization, gradually narrowing the gap between the obtained solution and the optimal solution. The PAH algorithm is designed to be adaptable, allowing to start from any feasible point. This flexibility makes it an ideal choice for application following the utilization of conventional heuristics.

Given the inherent complexity of the bundle pricing problem, which persists even when considering specific distributions (Babaioff et al., 2020), we compare the performance of PAH algorithm both analytically and numerically relative to simpler heuristics in the literature. It is important to note that while profitability of bundle pricing was recognized in research as early as 60 years ago by Stigler (1963), the extent to which it can improve profits relative to the case of no-bundling (component pricing) has only recently been thoroughly explored (Hart, Nisan, et al., 2013; Hart & Nisan, 2017). The use of pure bundling as a heuristic for bundle pricing has been widely adopted since the early stages of introducing bundle pricing as an effective marketing strategy. However, recently, its worst-case guarantees for additive independent reservation prices have been established (for details, see Babaioff et al. (2020), and Ma and Simchi-Levi (2021)). Furthermore, despite the introduction of bundle size pricing as an approximation for bundle pricing in the previous decade by Hitt and Chen (2005), there is currently a lack of evidence demonstrating its effectiveness in approximating the optimal bundle pricing solution, even for specific reservation prices, except for the asymptotic analysis.

We also introduce an extended version of PAH called the extended pricing-assortment heuristic (E-PAH) to further enhance the performance of the PAH. The E-PAH reduces dependence on the initial starting point and iteratively improves upon the best solution obtained from the PAH, leading to increased profits and decreased reliance on the starting point. Comprehensive numerical analysis are conducted to evaluate the efficacy of the proposed heuristics under various conditions. The results demonstrate the superiority of the offered heuristics compared to existing approaches, even when considering products with complex interrelationships. This study provides a practical and effective approach to the bundling problem, which can be applied across a wide range of industries, enabling businesses to increase their profits through bundle pricing strategies.

The remaining sections of the paper are structured as follows: Section 2 offers a comprehensive review of the relevant literature. Section 3 presents the formulation of the bundling problem. Section 4 elaborates on PAH algorithm, while Section 5 assesses its performance analytically. Furthermore, Section 6 provides an E-PAH. In Section 7, we employ numerical studies to demonstrate the superiority of our heuristics. Finally, Section 8 encapsulates our conclusions and outlines potential avenues for future research.

2 Literature Review

The exploration of bundling began with the concept of mixed bundling, where two products as well as bundles are offered (Adams & Yellen, 1976; Stigler, 1963). Extensive literature has emphasized the advantages of bundle pricing. For instance, Cao et al. (2019) discussed bundling as a way of creating a balance between supply and demand, while C. Wu et al. (2022) illustrated that bundling can be an efficient strategy in the market with new products if consumers' search cost is not high. These studies specifically addressed the two-product bundling scenarios.

The classical two-product model has garnered significant attention due to the persistent computational complexity of the bundling problem, even when consumer valuations are discrete (Hanson & Martin, 1990). It is noteworthy that the NP-hard nature of the problem poses challenges in finding optimal solutions. Consequently, researchers have proposed heuristic approaches to tackle this complexity, which can be broadly classified into two main groups.

The first group is pure bundling, but, this strategy often raises concerns regarding legality under the per se rule and the rule of reason (Stremersch & Tellis, 2002). Although, Dillbary (2010) demonstrated that mixed bundling can increase social welfare even at a price lower than its cost. Furthermore, it can result in low profits when the marginal cost of products is not relatively low and the variance between customers utilities is not small enough (Abdallah, 2019; Stremersch & Tellis, 2002). Furthermore, a study conducted by Derdenger and Kumar (2013) that examined bundling effects in a video market for five years found that pure bundling reduces profits by 20% compared with mixed bundling and reduces both hardware and software sales by millions. According to a report by Forbes (Gerdeman, 2013), bundling is frequently successful only if the consumer has the option of purchasing some of the products separately or as a part of another bundle.

The second group is cardinality bundling, also known as bundle size pricing, which allows customers to choose any combination of products, and the price they pay is dependent only on the number of products, not the selection of products. Chu et al. (2011) demonstrated that bundle size pricing can approximate mixed bundling through numerical analysis, and it is more profitable than component pricing. However, due to increasing complexity, they considered only five products. Solving a bundle size pricing is typically challenging or even impossible for large number of products, except in very exceptional circumstances, due to the need for order statistics (Briskorn et al., 2016). Moreover, bundle size pricing is not efficient when the marginal costs of the products are heterogeneous (Abdallah et al., 2021), or when the value of bundles of a given size varies widely among customers. Therefore, it should be used with caution, unless it offers a clear advantage over simpler approaches.

We enhance upon previous heuristics by offering a wider range of prices for bundles while also relaxing some of their restrictive assumptions.

3 Model Description

Bundle pricing is essentially a strategy to determine which bundles should be offered, and at what price. It is therefore a combination of assortment and pricing. The joint assortment and pricing corresponding to an optimal bundling policy has an exponentially large number of binary and continuous variables with lack of practical solution. Therefore, we seek an algorithm that iteratively analyzes assortment and pricing and helps to limit the number of variables and complexity of the problem. In this section, we present a formulation of the bundle pricing problem, and its relation to pricing and assortment optimization. We will also provide a high-level overview of our heuristic approach to address this problem.

3.1 Notation

Scalars will be denoted by lower case letters, vectors by bold letters, matrices by bold capital letters, and sets by capital letters. We will slightly abuse notation by using capital letters to refer to the seller's profit. Additionally, to distinguish parameters from variables in optimization problems, we denote decision variables by the tilde notation. For brevity of notation, we drop the tilde notation when there is no possibility of misunderstanding.

Throughout this research, we may use the terms "bundle pricing" and "mixed bundling" interchangeably, as they pertain to closely related concepts.

3.2 Formulation of Mixed Bundling Problem

Let $N := \{1, 2, ..., n\}$ be the set of products. We represent subsets of N in vector form: $\boldsymbol{x} \in \{0, 1\}^n$. Associated with each bundle $\boldsymbol{x} \neq \mathbf{0}$ there is a price $p(\boldsymbol{x}) \geq 0$, and cost $c(\boldsymbol{x})$. The bundle $\mathbf{0}$ representing not-purchasing, is assumed to always be available at a zero cost and price; i.e., $c(\mathbf{0}) = p(\mathbf{0}) = 0$. The set of bundles with finite prices are those that are offered to consumers, who are assumed to be heterogeneous and have unknown types. Arriving consumers can be grouped into a finite set I, where each type $i \in I$ is assumed to be homogeneous. The probability that a consumer belongs to a particular type i is denoted by α_i, and the set of all probabilities is represented as $\boldsymbol{\alpha} := (\alpha_1, ..., \alpha_{|I|})$. The consumers type i has a reservation price $v_i(\boldsymbol{x}) \geq 0$ for bundle $\boldsymbol{x}$. This reservation price is the highest price that a consumer in type i is willing to pay for the bundle, and thus serves as an indicator of the consumer's valuation of the product. This valuation is normalized so that the empty set has a reservation price zero for all types; i.e., $v_i(\mathbf{0}) = 0$ for all $i \in I$.

Our approach to the mixed bundling problem is focused on direct mechanism design, which entails formulating the problem as a bundle assignment and payment rule. In order to ensure the mechanism is incentive-compatible, it must be designed such that misreporting does not increase the surplus of any consumer. Additionally, the mechanism must be individual rational, meaning that participation leads to a positive surplus for each participant. To provide a more precise formulation of the problem, we introduce an indicator variable $y_i(\boldsymbol{x})$ which takes the value of 1 if a consumer in type i prefers bundle $\boldsymbol{x}$ over all other bundles with finite prices, and 0 otherwise. The formulation of the mixed bundling (MB) problem with this mechanism, as proposed by Spence (1980), is as follows:

$$
\begin{aligned}
OPT_{MB} = \max_{\tilde{y}_i(\boldsymbol{x}),\, \tilde{p}(\boldsymbol{x})} & \sum_{i \in I} \sum_{\boldsymbol{x}} \alpha_i \tilde{y}_i(\boldsymbol{x})(\tilde{p}(\boldsymbol{x}) - c(\boldsymbol{x})) \\
& s.t. \sum_{\boldsymbol{x}} \tilde{y}_i(\boldsymbol{x})(v_i(\boldsymbol{x}) - \tilde{p}(\boldsymbol{x})) \geq v_i(\boldsymbol{x}^{'}) - \tilde{p}(\boldsymbol{x}^{'}) \quad \forall i \in I, \boldsymbol{x}^{'} \in \{0,1\}^n \\
& \sum_{\boldsymbol{x}} \tilde{y}_i(\boldsymbol{x}) \leq 1 \ \forall i \in I \\
& \tilde{y}_i(\boldsymbol{x}) \in \{0,1\} \ \forall i \in I, \boldsymbol{x} \in \{0,1\}^n \\
& \tilde{p}(\boldsymbol{x}) \geq 0 \ \forall \boldsymbol{x} \in \{0,1\}^n
\end{aligned} \tag{MB}
$$

The first constraint satisfies individual rationality, since the right hand includes $\mathbf{0}$ which is priced at 0 by default, ensuring a non-negative surplus from the purchase. It also satisfies incentive compatibility by ensuring that the chosen bundle provides a surplus at least as high as any other.

The (MB) problem is mixed integer programming with $2^n \times |I|$ binary variables, and $2^n - 1$ real variables. It is well known that mixed integer programming is generally NP-complete. There are, however, instances in which the (MB) problem can be solved in polynomial time, particularly when $n = 1$. For a single product, Myerson (1981) illustrated there is an optimal deterministic mechanism. This polynomial time solution can be used as a heuristic algorithm for the bundling problem with a slight modification. In particular, the (MB) problem can be approximated by solving separate pricing problems for each bundle using Myerson's mechanism and then selecting the bundles that maximize profit, i.e., fix $p(\boldsymbol{x})$s at Myerson's mechanism and solve an assortment optimization problem. In the following, we will discuss how this approach can result in a constant factor approximation for the scenario where reservation prices are independent and additive. Independent reservation prices refer to situations where there is no correlation between the reservation prices of different products, while additive reservation prices refer to cases where the reservation price for a bundle is simply the sum of the reservation prices of its components.

We define $\boldsymbol{e}_n$ as a vector in $\{0,1\}^n$ with a value of 1 in the nth position and 0s elsewhere. Let $\boldsymbol{p} = (p(\boldsymbol{e}_1), ..., p(\boldsymbol{e}_n))$ be a vector of prices for individual products. Under the component pricing strategy, bundle prices are determined as the sum of their component prices. In other words, the firm selects a price vector $\boldsymbol{p}$, and consumers can freely choose any bundle $\boldsymbol{x}$ at a price of $\boldsymbol{x}^T\boldsymbol{p}$. When independent and additive reservation prices for products are assumed, Myerson's mechanism can be applied to the component pricing. This means that the optimal solution for component pricing with independent and additive reservation prices is to price products using Myerson's mechanism.

Furthermore, pure bundling, which refers to offering only the grand bundle of all products, can be seen as a case with one product, so the Myerson's mechanism can be used to find its optimal prices.

Therefore, solving the Myerson's mechanism for each bundle and then solving an assortment optimization by fixing these prices produces a revenue higher than both component pricing and pure bundling for the case with independent additive reservation prices. Since the seller can only offer the grand bundle or offer single products, and consumers can purchase any combination of the products.

On the other hand, for the case of independent and additive reservation prices, Ma and Simchi-Levi (2021) demonstrated that the maximum of component pricing and pure bundling provides a $\frac{1}{5.2}$-approximation for mixed bundling, which is the best known bound. Therefore, in this special case, the combination of pricing and assortment optimization guarantees to achieve at least $1/5.2$ of the optimal mixed bundling strategy, making it a suitable heuristic method for mixed bundling.

However, incorporating correlated and non-additive reservation prices into the problem can eliminate the worst-case guarantees for mixed bundling, and finding a non-zero guarantee for the general case is NP-hard (Hart, Nisan, et al., 2013). As a result, there have been limited studies that consider non-additive reservation prices, and those that do often rely on another restrictive assumption: the monotonicity of consumers' reservation prices (see for e.g. Ghili (2023)). This assumption asserts that

$$v_i(\boldsymbol{x}) > v_i(\boldsymbol{x}^{'}) \Rightarrow v_j(\boldsymbol{x}) > v_j(\boldsymbol{x}^{'})$$

Nonetheless, in reality, products frequently exhibit intricate relationships, and the preferences for bundle order can vary among consumers. To address this complexity and overcome the lack of practical heuristics, we propose heuristics for bundle pricing that are independent of product relationships. Building on the efficiency of the combination of pricing and assortment optimization, we introduce algorithms that alternate between pricing and assortment optimization.

4 Pricing-Assortment Heuristic

The purpose of this section is to present the pricing-assortment heuristic (PAH). The offered algorithm can start with any feasible solution and iterate between price optimization for fixed bundles and bundle optimization for fixed prices. The objective function monotonically improves with each iteration. One practical application of this profit improvement strategy is the introduction of consultants whose purpose is to improve company performance and increase profits.

The proposed heuristics can be applied to the bundle pricing problem regardless of the consumer's reservation price. This feature enables us to use a standard transformation technique to convert the bundle pricing problem into one with zero marginal cost.

$$v_i(\boldsymbol{x}) \leftarrow v_i(\boldsymbol{x}) - c(\boldsymbol{x})$$
$$p(\boldsymbol{x}) \leftarrow p(\boldsymbol{x}) - c(\boldsymbol{x})$$

As a result, the updated $v_i(\boldsymbol{x})$ may be negative, or non-increasing.

4.1 The Pricing-Assortment Heuristic

Pricing step. Given any feasible assignment, $y_i(\boldsymbol{x})$, $\forall i \in I$ and $\boldsymbol{x} \in \{0,1\}^n$, let $\boldsymbol{x}(i) := \{\boldsymbol{x} \in \{0,1\}^n : y_i(\boldsymbol{x}) = 1\}$ denote the bundle selected by consumer type $i \in I$, and let $\hat{I} := \{i \in I : \boldsymbol{x}(i) \neq \mathbf{0}\}$ be the set of consumer types for which it is optimal to buy a non-trivial bundle. The seller's profit from these consumers is $\sum_{i \in \hat{I}} \alpha_i p(\boldsymbol{x}(i))$. The goal of this step is to optimize over prices so that the consumers in set $\hat{I}$ continue to select the same bundles. This gives rise to the following linear programming (LP) formulation:

$$
\begin{aligned}
OPT_{LP}(\boldsymbol{Y}) := \max_{\tilde{p}(\boldsymbol{x})} \sum_{i \in \hat{I}} \alpha_i \tilde{p}(\boldsymbol{x}(i)) \\
s.t. v_i(\boldsymbol{x}(i)) - \tilde{p}(\boldsymbol{x}(i)) \geq v_i(\boldsymbol{x}(j)) - \tilde{p}(\boldsymbol{x}(j)) \quad \forall i, j \in \hat{I} \\
v_i(\boldsymbol{x}(i)) \geq \tilde{p}(\boldsymbol{x}(i)) \quad \forall i \in \hat{I} \\
\tilde{p}(\boldsymbol{x}(i)) \geq 0 \quad \forall i \in \hat{I},
\end{aligned}
\tag{LP}
$$

Where $\boldsymbol{Y}$ refers to the assignment matrix. The first and second set of constraints ensure that the bundle assignment to consumers who select a non-zero bundle remains unchanged from the initial feasible point. The following proposition illustrates that the LP problem has a unique solution. All proof can be found in the Appendix.

Proposition 1. *The program (LP) has a unique bounded solution that is independent of α_i, $i \in I$.*

This proposition is based on the intuition that, an optimal solution for different $\boldsymbol{\alpha}$ is always feasible since constraints are independent of $\boldsymbol{\alpha}$. Moreover, the maximum of two feasible solutions is also a feasible solution since the first constraint is equivalent to

$$
\tilde{p}(\boldsymbol{x}(i)) \leq v_i(\boldsymbol{x}(i)) - v_i(\boldsymbol{x}(j)) + \tilde{p}(\boldsymbol{x}(j)) \tag{1}
$$

The (LP) problem is used to determine the prices of bundles that are purchased by at least one consumer type. However, for bundles that have not yet been purchased, a different approach is taken to determine their price. Specifically, the price $p(\boldsymbol{x})$ for such bundles $\boldsymbol{x}$ is calculated using the formula $p(\boldsymbol{x}) = \max\{v_i(\boldsymbol{x}) - (v_i(\boldsymbol{x}(i)) - p(\boldsymbol{x}(i)) : i \in \hat{I}\}$, where $p(\boldsymbol{x}(i))$ is the solution obtained from the LP. It is important to note that these prices do not affect the current selection of consumers in $\hat{I}$.

Assortment step. The objective of this step is to assign the bundle with the highest profit contribution to consumer types without changing their surpluses. Let $p(\boldsymbol{x})$ be the updated vector of prices given by pricing step. Fixing these prices, the following assortment optimization problem denoted by

(ASST) is solved, remembering that costs have been absorbed into $p(x)$ and $v_i(x)$.

$$
\begin{aligned}
OPT_{ASST}(\boldsymbol{P}) = \max_{\tilde{y}_i(\boldsymbol{x})} & \sum_{\boldsymbol{x}} \sum_{i \in I} \alpha_i \tilde{y}_i(\boldsymbol{x}) p(\boldsymbol{x}) \\
s.t. & \sum_{x} \tilde{y}_i(\boldsymbol{x})(v_i(\boldsymbol{x}) - p(\boldsymbol{x})) \geq v_i(\boldsymbol{x}^{'}) - p(\boldsymbol{x}^{'}) \quad \forall i \in I, \boldsymbol{x}^{'} \in \{0,1\}^n \\
& \sum_{\boldsymbol{x}} \tilde{y}_i(\boldsymbol{x}) \leq 1 \quad \forall i \in I \\
& \tilde{y}_i(\boldsymbol{x}) \in \{0,1\} \quad \forall i \in I
\end{aligned} \tag{ASST}
$$

Here, $\boldsymbol{P}$ denotes the prices matrix obtained from the pricing step. The optimization problem in (ASST) is a linear binary programming problem over $y_i(\boldsymbol{x})$s. Although the objective function of the optimization problem is formulated as a seller optimization problem, with $\sum_{\boldsymbol{x}} \sum_{i \in I} \alpha_i \tilde{y}_i(\boldsymbol{x}) p(\boldsymbol{x})$ serving as the seller's objective function, it is important to note that the decision variables $y_i(\boldsymbol{x})$s are actually associated with consumers, rather than the seller. This is due to the consumer-oriented nature of the constraint, which remains valid even in the context of the (MB) problem. As a result, purchasing decisions are ultimately made by the consumers themselves, through the incentive compatibility constraint. Furthermore, as the seller lacks prior knowledge regarding the types of arriving consumers, the selected set of bundles will be offered to all consumers, and the assortment offered to them consists of bundles for which there exists $i \in I$ such that $y_i(\boldsymbol{x}) = 1$.

The problem (ASST) can be decomposed into $|I|$ separate binary programs and can be solved in $O(|I|)$ time. As the objective function is maximization with positive coefficients for all variables, and constraints can be categorised in terms of consumers, with each constraint containing only variables related to one type of consumer. Therefore, the assignment problem can be solved independently for each consumer type in $O(1)$ time. To see how this is done, consider the problem for type i.

$$
\begin{aligned}
OPT_{ASST_i}(\boldsymbol{P}) = \max_{\tilde{y}_i(\boldsymbol{x})} & \sum_{\boldsymbol{x}} \tilde{y}_i(\boldsymbol{x}) p(\boldsymbol{x}) \\
s.t. & \sum_{\boldsymbol{x}} \tilde{y}_i(\boldsymbol{x})(v_i(\boldsymbol{x}) - p(\boldsymbol{x})) \geq v_i(\boldsymbol{x}^{'}) - p(\boldsymbol{x}^{'}) \quad \forall \boldsymbol{x}^{'} \in \{0,1\}^n \\
& \sum_{\boldsymbol{x}} \tilde{y}_i(\boldsymbol{x}) \leq 1 \\
& \tilde{y}_i(\boldsymbol{x}) \in \{0,1\}
\end{aligned} \tag{ASST$_i$}
$$

Let $su(i) := \max_{\boldsymbol{x}} \{v_i(\boldsymbol{x}) - p(\boldsymbol{x})\}$ be the maximum surplus that can be obtained by consumers type i. There may be one or more bundles that achieve surplus $su(i)$. For each such bundle the profit contribution is $p(\boldsymbol{x})$, so while consumers are indifferent among all bundles with surplus $su(i)$, the firm is not, so the optimization will select the surplus maximizing bundle that also maximizes the profit contribution. This may require slightly modifying prices, to make it incentive-compatible for consumers to select the profit maximizing bundle. This adjustments need to be made carefully to avoid conflicts for different types.

Of course this is only a problem where there are multiple bundles that have the same surplus.

Algorithm 1 Pricing-Assortment Heuristic

Input: $v_i(\boldsymbol{x})$, $\boldsymbol{\alpha}$, a feasible point $\boldsymbol{x}(i)$ as an initial, $\forall i \in I, \boldsymbol{x} \in \{0,1\}^n$
1: $R^0 = 0$
2: $\boldsymbol{P}$ = price optimization ($\boldsymbol{Y}$)
3: $R = \sum_{i \in I} \alpha_i p(\boldsymbol{x}(i))$
4: While $R^0 < R$:
    4.1: $R^0 = R$
    4.2: $\boldsymbol{P}$ = price optimization ($\boldsymbol{Y}$)
    4.3: $\boldsymbol{Y}$, R=assortment optimization ($\boldsymbol{P}$)
Output: $\boldsymbol{P}$, $\boldsymbol{Y}$, and R

// Pricing and assortment optimization functions
function price optimization ($\boldsymbol{Y}$)
    Solve LP and find $p(\boldsymbol{x}(i)) \ \forall \, i \in \hat{I}$
    $p(\boldsymbol{x}) \leftarrow \max\{v_i(\boldsymbol{x}) - (v_i(\boldsymbol{x}(i)) - p(\boldsymbol{x}(i)) : i \in \hat{I}\}$
    return $\boldsymbol{P}$
end function
function assortment optimization ($\boldsymbol{P}$)
    for each, $i \in I$ **do**
        $su(i) \leftarrow \max_{\boldsymbol{x}}\{v_i(\boldsymbol{x}) - p(\boldsymbol{x})\}$
        $\boldsymbol{x}(i) \leftarrow \arg\max_{\boldsymbol{x}}\{p(\boldsymbol{x}) : v_i(\boldsymbol{x}) - p(\boldsymbol{x}) = su(i)\}$
        $\hat{I} \leftarrow \{i \in I : \boldsymbol{x}(i) \neq \boldsymbol{0}\}$
    end for
    $R \leftarrow \sum_{i \in \hat{I}} \alpha_i p(\boldsymbol{x}(i))$
    return $\boldsymbol{Y}$, R
end function

An optimal solution to (ASST$_i$) is to set $y_i(\boldsymbol{x}) = 1$, for a bundle with the highest price that has surplus su_i, and $y_i(\boldsymbol{x}) = 0$ for all other bundles. That is

$$y_i(\boldsymbol{x}) = \begin{cases} 1 & \boldsymbol{x} = \arg\max\{p(\boldsymbol{x}) : v_i(\boldsymbol{x}) - p(\boldsymbol{x}) = su(i)\} \\ 0 & \text{others} \end{cases}$$

Accordingly, we update $\boldsymbol{x}(i)$ and proceed to pricing step. These steps are repeated until no improvement can be achieved.

In order to demonstrate the functioning of the PAH algorithm, Algorithm 1 presents its corresponding pseudo-code. Furthermore, a detailed explanation of the individual steps is provided through the following example.

Example 1. *Consider a case with* 3 *consumer types and* 4 *products. The reservation price of consumers and the proportion of each type is shown in Table 1. Figure 1 outlines the steps in the PAH algorithm for this scenario.*

	Consumer 1	Consumer 2	Consumer 3
(1,0,0,0)	28	12	29
(0,1,0,0)	74	60	48
(0,0,1,0)	53	98	52
(0,0,0,1)	7	69	33
(1,1,0,0)	78	63	76
(1,0,1,0)	56	110	75
(1,0,0,1)	28	73	38
(0,1,1,0)	112	106	66
(0,1,0,1)	79	122	70
(0,0,1,1)	58	145	58
(1,1,1,0)	114	130	82
(1,1,0,1)	90	128	107
(1,0,1,1)	59	156	90
(0,1,1,1)	118	180	90
(1,1,1,1)	136	189	158
Consumer Probabilities	0.1835	0.4427	0.3738

Table 1: Consumers reservation prices and probabilities

Assignment Initial Point	Consumer 1	Consumer 2	Consumer 3
	(0,0,0,0)	(1,1,1,1)	(0,0,0,0)

Pricing Step

Prices First Iteration	(1,0,0,0)	(0,1,0,0)	(0,0,1,0)	(0,0,0,1)	(1,1,0,0)	(1,0,1,0)	(1,0,0,1)	(0,1,1,0)	(0,1,0,1)	(0,0,1,1)	(1,1,1,0)	(1,1,0,1)	(1,0,1,1)	(0,1,1,1)	(1,1,1,1)
	12	60	98	69	63	110	73	106	122	145	130	128	156	180	189

Assortment Step

Assignment First Iteration	Consumer 1	Consumer 2	Consumer 3
	(1,1,0,0)	(1,1,1,1)	(1,0,0,0)

Pricing Step

Prices Second Iteration	(1,0,0,0)	(0,1,0,0)	(0,0,1,0)	(0,0,0,1)	(1,1,0,0)	(1,0,1,0)	(1,0,0,1)	(0,1,1,0)	(0,1,0,1)	(0,0,1,1)	(1,1,1,0)	(1,1,0,1)	(1,0,1,1)	(0,1,1,1)	(1,1,1,1)
	29	74	98	69	78	110	73	112	122	145	130	128	156	180	189

Assortment Step

Assignment Second Iteration	Consumer 1	Consumer 2	Consumer 3
	(0,1,1,0)	(1,1,1,1)	(1,0,0,0)

Figure 1: Implementing PAH algorithm

Figure 1 provides the necessary data to calculate the seller's profits, which undergo a remarkable improvement through the PAH algorithm. Specifically, the expected profits increase from 83.9 *to* 115 *after two iterations, with the seller's profits reaching* 99.9 *after the first iteration and* 115 *after the second iteration. The algorithm terminates after the second iteration since no further improvement is obtained, resulting in a* 37% *increase in profits. Importantly, the initial point assignment restricts the expected profits to no more than* 83.9 *because the prices that maximize profits remain the same as those in the first iteration.*

We will now demonstrate the convergence of the algorithm with the following propositions.

Proposition 2. *The price of bundles weakly increases in each iteration through PAH algorithm.*

As a result, the consumer surplus weakly decreases in each iteration, and due to the individual rationality constraint, the algorithm converges. The next proposition establishes that if all consumers have zero surplus, the PAH algorithm terminates.

Proposition 3. *The PAH algorithm ensures that at least one consumer type is charged at their reservation price. Moreover, if the algorithm extracts all consumer surplus, it will terminate, indicating that repeating it would not yield any further profit enhancement.*

5 Analyzing Potential Profit Improvement with the PAH

Given the NP-hardness of finding an approximation with a better worst-case guarantee than zero for the general case, as well as the inherent complexity of bundle pricing even for specific distributions (Babaioff et al., 2020; Hart, Nisan, et al., 2013), this section aims to compare the performance of the PAH algorithm with simpler alternatives. Specifically, the section aims to demonstrate the potential improvement in profit achievable by the PAH algorithm compared to common bundle pricing heuristics such as pure bundling and bundle size pricing, as well as starting from arbitrary solutions. We commence this section by demonstrating the ability of the PAH algorithm to enhance the expected seller profit in a general setting by initializing it from an arbitrary solution. Subsequently, in the ensuing subsection, we focus on the cases where pure bundling and bundle size pricing are employed as the initial point.

Let $v_{\max} := \max\limits_{i,\boldsymbol{x}} v_i(\boldsymbol{x})$ and $v_{\min} := \min\limits_{i,\boldsymbol{x}} \{v_i(\boldsymbol{x}) : v_i(\boldsymbol{x}) > 0\}$ denote the maximum and minimum reservation prices, respectively, over all consumers and possible bundles. Furthermore, let a represent an arbitrary feasible initial solution, and let R_a represent the expected profits associated with a. After applying the PAH algorithm with the initial point a, let $R_{P,a}$ denote the resulting expected profits. We aim to demonstrate that the PAH algorithm can achieve significant profit improvement starting from any feasible initial solution a.

Proposition 4. *For uniformly distributed $\alpha_i = \frac{1}{|I|}$, $i \in I$,*

$$R_{P,a} \leq \frac{|I| v_{\max}}{v_{\min}} R_a$$

Moreover the bound is tight.

For non-uniform distributions, the ratio $\frac{R_{P,a}}{R_a}$ can be arbitrarily larger than $\frac{|I| v_{\max}}{v_{\min}}$, as shown in the following example.

Example 2. *Assume a case with two products such that $v_i(0,1) = 0$, $\forall i \neq j$, $v_j(0,1) = \epsilon = \alpha_j$, $v_i(1,0) = 0.4$, $\forall i \neq l$, $v_l(1,0) = 0.49 = v_l(1,1)$, and $v_i(1,1) = 1$, $\forall i \neq l$. Selling product 2 at a price ϵ and earning a profit of ϵ^2 is a feasible starting point. However, by implementation of PAH algorithm, product 1 at price* 0.49 *and a grand bundle of two products at a price of* 1 *will be offered,* *resulting in a profit of* $(1-0.51\alpha_l)$. *Thus:* $\frac{R_{P,a}}{R_a} = \frac{1-0.51\alpha_l}{\epsilon^2}$. *On the other hand,* $\frac{|I|v_{\max}}{v_{\min}} = \frac{|I|}{\epsilon}$. *Hence:* $\frac{R_{P,a}/R_a}{|I|v_{\max}/v_{\min}} = \frac{1-0.51\alpha_l}{|I|\epsilon} \to \infty$

As a result, the PAH algorithm represents a powerful tool for amplifying seller profitability by expanding product bundles, enabling higher pricing and the extraction of full surplus from consumers. However, offering a single bundle $(0,1)$ fails to account for the significant benefits that can be obtained from consumer heterogeneity. The PAH algorithm identifies this inefficiency and proposes the provision of two distinct bundles, each targeted different consumer types. Specifically, the seller offers a mixed bundle comprising solely of product 1 at a premium price for type-l consumers, who characterized by a higher reservation price for product 1 but a lack of valuation for the bundle with product 2, and the grand bundle for all other consumer types. This socially efficient solution generates substantial profitability for the seller, highlighting the numerous advantages of the PAH algorithm.

Furthermore, it is worth noting that despite the violation of the Spence-Mirrlees single crossing condition, the algorithm is still capable of providing analytical outcomes. This condition, which assumes that consumers can be ranked based on their bundle valuations with higher valuations corresponding to higher marginal changes, is the main assumption in most studies that derive analytical results (see, e.g, Spence (1980) and J. Wu et al. (2019)). Therefore, the PAH algorithm's ability to provide analytical outcomes, despite violating this condition, is a testament to its strength and versatility.

5.1 Starting From Pure Bundling

In this subsection, we will explore the profit improvement achieved by the PAH algorithm when starting from an optimal solution of pure bundling. As a recall, pure bundling refers to the bundling strategy where only the grand bundle of all products is offered. We will begin by examining the general case and then move on to the scenario with zero marginal cost.

Lemma 1. *Given the optimal solution of pure bundling as the initial point and considering general reservation prices and non-zero costs, the PAH algorithm can improve seller profits arbitrarily.*

The example used to prove the lemma shows that the PAH algorithm can arbitrarily improve seller profit even when considering additive reservation prices. This finding is consistent with existing literature (see, for example, Bakos and Brynjolfsson (1999), Abdallah (2019)) that suggests that increasing the cost, the pure bundling can lead to arbitrarily poor outcomes. However, the PAH algorithm is less sensitive to cost and therefore outperforms pure bundling in this regard.

In the following proposition, we will evaluate the profit improvement potential of the PAH algorithm, considering zero cost. We denote the expected profit from pure bundling as R_u, and the expected profit after implementing the PAH algorithm following pure bundling as $R_{P,u}$.

Proposition 5. *Starting from an optimal solution of pure bundling and assuming zero cost, the PAH* *algorithm can increase expected profits up to*

$$R_{P,u} \leq R_u \sum_{i \in I} \frac{1}{i} < R_u(\ln(|I|) + 1)$$

In the proof of the proposition, we demonstrate that non-homogeneous reservation prices of consumers can result in poor performance of pure bundling. However, the PAH algorithm, by providing a larger range of products, can significantly increase profits. This is particularly important because consumers may have significantly varying reservation prices for different bundles.

It is worth emphasizing that the aforementioned result is not in conflict with the theoretical findings of Bakos and Brynjolfsson (1999), who demonstrated that pure bundling is an asymptotically optimal bundling strategy. Their theoretical result relies on the assumption of homogeneous consumer reservation prices, which permits the use of the central limit theorem. Nevertheless, as noted by S. Wu et al. (2008), considering homogeneous reservation prices is equivalent to the single consumer case problem.

5.2 Starting From Bundle Size Pricing

This subsection compares the performance of the bundle size pricing and PAH algorithms. In particular, we evaluate the extent to which the PAH can increase seller profits, considering the optimal solution of bundle size pricing as a starting point.

Let R_b denote the seller's profit from bundle size pricing, and $R_{P,b}$ denote the seller's profit from PAH algorithm when considering optimal solution of bundle size pricing as an initial point. Considering zero cost, Proposition 5 reveals that the improvement of the PAH algorithm relative to bundle size pricing is upper bounded by $R_b(\ln(|I|) + 1)$, given that pure bundling is a special case of bundle size pricing where all bundle sizes except the grand bundle have infinite prices. While this bound is the upper bound of a harmonic series and therefore not tight, it raises the question of whether there exists an instance of bundle size pricing where the performance of the PAH algorithm relative to bundle size pricing is the harmonic series. The following corollary illustrate that $R_{P,b} \leq R_b \sum_{i \in I} \frac{1}{i}$ is tight bound.

Corollary 1. *Assuming zero cost, the profit of the PAH algorithm, beginning from the optimal solution of BSP, is upper bounded by:*

$$R_{P,b} \leq R_b \sum_{i \in I} \frac{1}{i} < R_b(\ln(|I|) + 1)$$

6 The Extended Pricing-Assortment Heuristic

The PAH algorithm has the potential to significantly increase profits, however its performance is highly dependent on the initial point. Proposition 2 highlights that bundle prices in the PAH algorithm can only be adjusted upwards, indicating that the marginal change associated with each bundle price is non-negative. Therefore, if nothing has been assigned to a consumer type in the first iteration, nothing will be assigned in subsequent iterations. To mitigate the dependency on the initial point, we propose an algorithm called the extended pricing-assortment heuristic (E-PAH) in this section, which incorporates an additional step.

The E-PAH algorithm starts with the PAH algorithm and, upon convergence, includes an additional step to reduce the influence of the initial point, called escaping from initial point step. The objective of this step is to enhance the seller's profit and reduce dependence on the initial point by adjusting bundle prices. In particular, some bundle prices may be reduced to attract new consumers. This decrease may induce other consumers to switch to the lower-priced bundle causing profits to drop. Thus, this would require adjusting other prices in the hope of increasing overall profits. To keep the process tractable, we add at most one bundle at a time with a focus on nudging consumers towards larger bundle sizes.

Escaping from initial point step. Let $k_{\min} := 1 + \min_{i \in I} |\boldsymbol{x}(i)|$, where $\boldsymbol{x}(i)$ is the last update of the bundle selected by consumers type i. If $k_{min} \leq n$, then for $k \in \{k_{min}, ..., n\}$ do the following; otherwise, the algorithm terminates.

Let $I_k^{'} := \{i \in I : |\boldsymbol{x}(i)| < k\}$ be the set of consumer types for whom it is optimal to purchase a bundle of size less than k. Let $X_k := \{\boldsymbol{x} : |\boldsymbol{x}| = k\}$, and $X_k^{'} := \{\boldsymbol{x} : \exists i \notin I_k^{'},\ \boldsymbol{x}(i) = \boldsymbol{x}\} \cup \{\boldsymbol{0}\}$. Our objective is to increase profits by making bundles within X_k more attractive to consumers in $I_k^{'}$ where possible, while ensuring that consumers $i \notin I_k^{'}$ maintain their current choices. This problem is referred to as price adjustment of size k and denoted by PA_k.

$$
\begin{aligned}
OPT_{PA_k} = \max_{\tilde{p}(\boldsymbol{x}),\, \tilde{y}_i(\boldsymbol{x})} & \sum_{i \in I_k^{'}} \sum_{\boldsymbol{x} \in X_k} \alpha_i \tilde{y}_i(\boldsymbol{x}) \tilde{p}(\boldsymbol{x}) + \sum_{i \notin I_k^{'}} \alpha_i \tilde{p}(\boldsymbol{x}(i)) \\
& s.t. v_i(\boldsymbol{x}(i)) - \tilde{p}(\boldsymbol{x}(i)) \geq v_i(\boldsymbol{x}^{'}) - \tilde{p}(\boldsymbol{x}^{'}) \qquad \forall i \notin I_k^{'},\ \boldsymbol{x}^{'} \in X_k \cup X_k^{'} \\
& \sum_{\boldsymbol{x} \in X_k} \tilde{y}_i(\boldsymbol{x})(v_i(\boldsymbol{x}) - \tilde{p}(\boldsymbol{x})) \geq v_i(\boldsymbol{x}^{'}) - \tilde{p}(\boldsymbol{x}^{'}) \qquad \forall i \in I_k^{'}, \boldsymbol{x}^{'} \in X_k \cup X_k^{'} \\
& \sum_{\boldsymbol{x} \in X_k} \tilde{y}_i(\boldsymbol{x})(v_i(\boldsymbol{x}) - \tilde{p}(\boldsymbol{x})) \geq su(i) \qquad \forall i \in I_k^{'} \\
& \sum_{\boldsymbol{x} \in X_k} \tilde{y}_i(\boldsymbol{x}) \leq 1 \qquad i \in I_k^{'} \\
& \tilde{y}_i(\boldsymbol{x}) \in \{0, 1\} \qquad \forall \boldsymbol{x} \in X_k, i \in I_k^{'} \\
& \tilde{p}(\boldsymbol{x}) \geq 0 \qquad \forall \boldsymbol{x} \in X_k \cup X_k^{'} \setminus \{\boldsymbol{0}\}
\end{aligned}
\tag{PA$_k$}
$$

The first set of constraints ensures that consumer types $i \notin I_k^{'}$ do not alter their choices. The third constraint guarantees that if a consumer type $i \in I_k^{'}$ opts to switch from their current selection to a larger bundle, they will be incentivized to do so as their surplus will increase.

Let $pp_i := \max_{\boldsymbol{x} \in X_k} v_i(\boldsymbol{x}) - su(i)$ indicate the maximum potential profit that can be gained by selling a bundle within size k to the consumer types $i \in I_k^{'}$.

Lemma 2. *In the optimal solution of problem* (PA$_k$)*, if* $pp_i \leq 0$*, then* $y_i(\boldsymbol{x}) = 0$*,* $\forall \boldsymbol{x} \in X_k$*.*

Accordingly, we consider only consumers in set $\{i \in I_k^{'} : pp_i > 0\}$. Although this may reduce the size of (PA$_k$), finding an exact solution can still be challenging. Thus, rather than allocating multiple bundles to multi consumers simultaneously, we focus on assigning one bundle to a consumer at a time. We start from consumer $i \in I_k^{'}$ who has a maximum pp_i, say j, we examine whether a profitable assignment can be made within X_k. Once we have checked a type of consumers, we remove them permanently from $I_k^{'}$, and they are no longer considered for allocation within this bundle size.

To speed up the algorithm, we select a bundle from X_k for consumers j that yields the greatest positive direct impact on profits, rather than exhaustively examining all bundles. We refer to $\alpha_j(v_j(\boldsymbol{x}) - su(j) - p(\boldsymbol{x}(j))) - \sum_{l \in I} \alpha_l \max\{0, v_l(\boldsymbol{x}) - su(l) - v_j(\boldsymbol{x}) + su(j)\}$ as a direct profit impact of assigning bundle $\boldsymbol{x} \in X_k$ to consumer type i. Following the update of consumer j's assignment, we repeat pricing and assortment steps (PAH algorithm) with the updated assignments. We then assess whether or not the revised assignment results in increased profits. If it does not, we reverse the price changes and restore the assignment of consumers to their previous status. This process is carried out for all consumers in $I_k^{'}$ with positive pp_i before moving on to the next bundle size. This procedure is described in more detail in Algorithm 2.

In Appendix B, we demonstrate numerically that escaping from initial point step significantly increases the average profit, while also reducing the standard deviation of the profits obtained from different initial points, thereby reducing dependence on the initial point. However, incorporating this additional step results in an increased running time. Therefore, businesses should consider their specific structure and the urgency of decision-making when deliberating between the PAH and E-PAH algorithms (For a more comprehensive understanding of the numerical results and intuitive insights, refer to Section 7).

In the following example we will demonstrate how to implement the escaping from initial point step using the Example 1.

Example 3. *We can see that in Example 1,* $k_{\min} = 2$*, and* $I_2^{'} = \{3\}$*. As shown in Figure 2, bundle* $(1, 1, 0, 0)$ *is a candidate assignment for consumer type 3 within size 2 since it has a maximum positive direct profit impact. The assignment is updated, and the new set of assignments is used as the starting point for PAH. After one iteration of the pricing and assortment steps, the problem converges, and profits increase to* 132.4*. The updated assignment in this case is the same as the assignment obtained before implementing PAH. It is worth noting that while PAH keeps the current assignment, it increases profits by changing prices.*

The algorithm is then applied to the next bundle size, $k = 3$*, with* $I_3^{'} = \{1, 3\}$*. Since there is no candidate bundle within size 3 for consumer type 1, who has maximum* pp_i*, it is removed from* $I_3^{'}$*. The algorithm identifies a candidate bundle for consumer type 3, say* $(1, 1, 0, 1)$*, and updates the assignment accordingly. After one iteration of pricing and assortment steps, the problem converges, and profits increase to* 135.8*. Notably, the new assignment for consumer type 3 also changes the assignment for*

Direct Profit Impact k=2	I''_2	(1,1,0,0)	(1,0,1,0)	(1,0,0,1)	(0,1,1,0)	(0,1,0,1)	(0,0,1,1)
	Consumer 3	17	2	-12	-11	-10	-27

Update Initial point for PAH, k=2, j=3	Consumer 1	Consumer 2	Consumer 3
	(0,1,1,0)	(1,1,1,1)	(1,1,0,0)

Prices k=2, j=3	(1,0,0,0)	(0,1,0,0)	(0,0,1,0)	(0,0,0,1)	(1,1,0,0)	(1,0,1,0)	(1,0,0,1)	(0,1,1,0)	(0,1,0,1)	(0,0,1,1)	(1,1,1,0)	(1,1,0,1)	(1,0,1,1)	(0,1,1,1)	(1,1,1,1)
	29	72	98	69	76	110	73	110	122	145	130	128	156	180	189

Assignmnet k=2, j=3	Consumer 1	Consumer 2	Consumer 3
	(0,1,1,0)	(1,1,1,1)	(1,1,0,0)

PAH

Direct Profit Impact k=3	I''_3	(1,1,1,0)	(1,1,0,1)	(1,0,1,1)	(0,1,1,1)
	Consumer 1	-6	-28	-65	-26
	Consumer 3	-25	3	-24	-39

Update Initial point for PAH, k=3, j=3	Consumer 1	Consumer 2	Consumer 3
	(0,1,1,0)	(1,1,1,1)	(1,1,1,0)

Prices k=3, j=3	(1,0,0,0)	(0,1,0,0)	(0,0,1,0)	(0,0,0,1)	(1,1,0,0)	(1,0,1,0)	(1,0,0,1)	(0,1,1,0)	(0,1,0,1)	(0,0,1,1)	(1,1,1,0)	(1,1,0,1)	(1,0,1,1)	(0,1,1,1)	(1,1,1,1)
	29	74	77	48	78	89	52	112	101	124	114	107	135	159	168

Assignmnet k=3, j=3	Consumer 1	Consumer 2	Consumer 3
	(1,1,1,0)	(1,1,1,1)	(1,1,0,1)

PAH

Direct Profit Impact k=4	I''_4	(1,1,1,1)
	Consumer 1	-11
	Consumer 3	5

Update Initial point for PAH, k=4, j=3	Consumer 1	Consumer 2	Consumer 3
	(1,1,1,0)	(1,1,1,1)	(1,1,1,1)

Prices k=4, j=3	(1,0,0,0)	(0,1,0,0)	(0,0,1,0)	(0,0,0,1)	(1,1,0,0)	(1,0,1,0)	(1,0,0,1)	(0,1,1,0)	(0,1,0,1)	(0,0,1,1)	(1,1,1,0)	(1,1,0,1)	(1,0,1,1)	(0,1,1,1)	(1,1,1,1)
	29	74	67	38	78	79	42	112	91	114	114	107	125	149	158

Assignmnet k=4, j=3	Consumer 1	Consumer 2	Consumer 3
	(1,1,1,0)	(1,1,1,1)	(1,1,1,1)

PAH

Figure 2: Implementing escaping from initial point step

Algorithm 2 Extended Pricing-Assortment Heuristic

Input: $v_i(\boldsymbol{x})$, $\boldsymbol{\alpha}$, a feasible point $\boldsymbol{x}(i)$ as an initial, $\forall i \in I, \boldsymbol{x} \in \{0,1\}^n$
1: $p(\boldsymbol{x}), R$=price optimization $(\boldsymbol{x}(i), \forall i)$
2: $\boldsymbol{x}(i)$, $p(\boldsymbol{x}), R$=assignment optimization $(p(\boldsymbol{x}), \forall \boldsymbol{x})$
3: **for** $k = k_{\min}, ..., n$ **do**
  $I_k^{'} \leftarrow \{i \in I : |\boldsymbol{x}(i)| < k\}$
  while $I_k^{'}$ is not empty **do**
    $I_k^{"} \leftarrow \{i \in I_k^{'} : pp_i > 0\}$
    if $I_k^{"}$ is not empty **then**
      $j \leftarrow \arg\max_{i \in I_k^{"}} pp_i$
      Remove j from $I_k^{'}$
      $\boldsymbol{x}_k^{'} \leftarrow \arg\max_{\boldsymbol{x} \in X_k} \{\alpha_j(v_j(\boldsymbol{x}) - su(j) - p(\boldsymbol{x}(j))) - \sum_{l \in I} \alpha_l \max\{0, v_l(\boldsymbol{x}) - su(l) - v_j(\boldsymbol{x}) + su(j)\}\}$
      if $\alpha_j(v_j(\boldsymbol{x}_k^{'}) - su(j) - p(\boldsymbol{x}(j))) - \sum_{l \in I} \alpha_l \max\{0, v_l(\boldsymbol{x}_k^{'}) - su(l) - v_j(\boldsymbol{x}_k^{'}) + su(j)\} \geq 0$ **then**
        $\boldsymbol{x}^0(j) \leftarrow \boldsymbol{x}(j), \quad p^0(\boldsymbol{x}) \leftarrow p(\boldsymbol{x}), \forall \boldsymbol{x}, \;\; R^0 \leftarrow R$
        $\boldsymbol{x}(j) = \boldsymbol{x}_k^{'}$
        $p(\boldsymbol{x})$=price optimization $(\boldsymbol{x}(i), \forall i \in I)$
        $\boldsymbol{x}(i), p(\boldsymbol{x}), R$=assignment optimization $(p(\boldsymbol{x}), \forall \boldsymbol{x})$
        if $R < R^0$ **then**
          $\boldsymbol{x}(j) = \boldsymbol{x}^0(j), \;\; p(\boldsymbol{x}) = p^0(\boldsymbol{x}), \forall \boldsymbol{x}, \;\; R = R^0$
        else
          $I_k^{'} \leftarrow \{i \in I_k^{'} : |\boldsymbol{x}(i)| < k\}$
        end if
      end if
    else
      Empty $I_k^{'}$
    end if
  end while
end for
Output: $p(\boldsymbol{x}), \forall \boldsymbol{x}, \;\; \boldsymbol{x}(i), \forall i \in I$, and R

consumer type 1 to a bundle with size 3, despite there being no candidate bundle for type 1 within size 3. The process is repeated for $k = 4$, and the algorithm increases profits to 150.

To sum up, the PAH algorithm increases the seller's profits from 83.9 *to* 115*, resulting in a* %37 *improvement. The addition of the "escaping from initial point" step in E-PAH further improves the profits to* 150*, which is a* 30% *increase compared to the PAH algorithm and a* 79% *increase compared to the initial assignment.*

7 Numerical Analysis

In this section, we conduct a comprehensive numerical analysis to assess the effectiveness of the proposed heuristics for bundle pricing. We compare their performance with that of existing heuristics available in the literature as well as component pricing. Given the computational complexity of determining the optimal mixed bundling solution, we evaluate the performance of each algorithm using an upper bound approach for bundle pricing. Specifically, we compare their performance with a benchmark referred to as the clairvoyant approach. The clairvoyant possesses perfect knowledge of consumers' reservation prices and can offer each consumer type i a bundle $\boldsymbol{x}_i = \arg\max\{v_i(\boldsymbol{x}) - c(\boldsymbol{x})\}$ at a price $v_i(\boldsymbol{x}_i)$.

We conduct a comparative analysis of the performance of our algorithms in various scenarios, taking into account both the distribution of reservation prices of consumers and the inter-relatedness of products. Initially, we consider a scenario where products possess additive reservation prices, refer to as additive cases. Although this assumption may be violated in a wide range of real-world scenarios, it serves as a fundamental assumption for many existing heuristics in the literature, enabling a fair comparison with other algorithms specifically designed for this scenario. Subsequently, we expand our investigation to encompass cases involving more intricate product inter-relatedness. In these scenarios, the reservation prices of bundles are determined by the reservation prices of their individual components, as well as the interplay between any pair of products within the bundle, refer to as symmetric cases. Finally, we demonstrate the performance of our algorithm in a broader range of scenarios, encompassing more general and realistic settings, refer to as general case.

We examine three combinations of parameters, denoted as $(\{n = 4, m = 10\}, \{n = 7, m = 15\}, \{n = 10, m = 20\})$, where $m = |I|$, and report the results for each test case across 100 replications. In order to demonstrate the robustness of our heuristics, we perform our numerical study using different distributions for the reservation prices. Specifically, we draw the reservation price for consumer type i from the uniform distribution $[0, \mu_i]$, lognormal distribution with parameters $(\mu_i, 1)$, truncated-normal distribution with parameters $(\mu_i, 1)$ truncated at zero, and exponential distribution with parameter $(1/\mu_i)$. These distributions are commonly employed in the bundling literature (Abdallah et al., 2021; Li et al., 2021). While we present the results pertaining to the uniform distribution in this section, the outcomes for the remaining distributions are provided in the Appendix A. By subjecting our heuristics to multiple distributions, we can evaluate their performance under various scenarios and assess their ability to handle variations in the underlying data. This comprehensive approach enables us to draw meaningful conclusions about the practical efficacy of our heuristics.

To compare the performance of our algorithms relative to component pricing, pure bundling, and bundle size pricing, we compare them with the following algorithms whenever it is possible:

1. Component Pricing (COM): We consider the prices of the products set to their optimal solution based on the Myerson's mechanism (Myerson, 1981). This serves as a benchmark for evaluating the performance of our algorithm relative to the no-bundling.

2. Pure Bundling with Disposal Cost (PBDC) Algorithm: We evaluate the pure bundling with disposal cost algorithm proposed by Ma and Simchi-Levi (2021). This algorithm addresses the negative impact of costs on pure bundling and serves as an upper bound for pure bundling.

3. Convex Optimization Heuristic for Bundle Size Pricing: We assess the convex optimization heuris-

tic for bundle size pricing proposed by Li et al. (2021), which we will refer to as LST[1]. The authors demonstrate that their algorithm outperforms other approximation methods for bundle size pricing.

By comparing the performance of our algorithm against these established approaches, we gain insights into its effectiveness and relative superiority.

To compare the performance of our heuristics with bundle size pricing in additive and symmetric cases, we assume a cost function solely dependent on the size of the bundle. Specifically, we consider a cost function of the form $c(k) = k\lambda a$, where $k \in [n]$ and λ takes on two values, namely $\lambda = 0$ and $\lambda = 0.5$. This form of the cost function is consistent with recent bundle size pricing literature, including studies by Abdallah et al. (2021) and Li et al. (2021). To determine the parameter a, we calculate the mean reservation price of all consumers for all bundles within size 1 and use it as a reference value. By adopting this approach, we can evaluate the performance of our heuristics in comparison to bundle size pricing under different cost structures. To make a meaningful comparison, we consider uniform consumer probabilities within additive and symmetric cases, which is consistent with their assumptions.

However, in the general case, since we are unaware of how to extend the LST algorithm for implementation in this specific scenario, we consider a more comprehensive situation where we assume that the products possess non-additive and heterogeneous costs. In this scenario, we further assume that the probability of consumer type i is given by $\frac{\nu_i}{\sum_{j \in I} \nu_j}$, where ν_i is a random variable drawn from a uniform distribution within the range of $[0, 1]$.

We utilize $R_.$ and $T_.$ to denote the expected profits and running time of each algorithm. Additionally, we consider the optimal solution of pure bundling as the starting point for PAH and E-PAH, unless stated otherwise.

7.1 Additive Reservation Prices

Obtaining and analyzing comprehensive information regarding consumer preferences for all possible product bundles presents a formidable challenge, particularly in markets with a large number of products. This task necessitates computing $v_i(\mathbf{x})$ for every $i \in I$ and every $\mathbf{x} \in \{0, 1\}^n$, resulting in an exponential number of queries ($O(2^n)$) for each consumer type. To mitigate this computational burden, one common assumption is the consideration of additive reservation prices. This assumption posits that consumers have reservation prices that can be aggregated additively, as observed in studies such as Bakos and Brynjolfsson (1999) and Li et al. (2021). Under this assumption, the reservation price for a bundle $\mathbf{x}$ is simply the sum of the reservation prices for its individual elements. By adopting this assumption, the number of required queries reduces to $O(n)$ per consumer type.

Here, we explore two cases: homogeneous and heterogeneous reservation prices. The homogeneous

[1] The convex optimization used in the LST algorithm is solved using global optimization provided by Scipy in Python

case is more prevalent in the literature, as it simplifies the development of algorithms to handle the scenario. In this case, we consider a reservation price distribution with $\mu_i = \mu = 0.5$ for all $i \in I$. For the heterogeneous case, we consider that the distribution of reservation prices vary across consumer types. Specifically, we set $\mu_j = \mu \frac{j}{m}$, where $\mu = 0.5$. This setting allows for heterogeneity in reservation prices among different consumer types while maintaining a proportional relationship with the overall average reservation price.

				Performance					Time in Second				
λ	n	m	Descriptive Statistics	R_COM	R_PBDC	R_LST	R_PAH	R_EPAH	T_COM	T_PBDC	T_LST	T_PAH	T_EPAH
0	4	10	mean	0.59	0.69	0.62	0.72	0.73	0.01	0.01	0.08	0.03	0.15
			std	0.04	0.05	0.06	0.05	0.06	0.00	0.00	0.01	0.00	0.08
	7	15	mean	0.57	0.72	0.65	0.75	0.77	0.01	0.01	0.18	0.04	0.36
			std	0.03	0.05	0.08	0.05	0.05	0.00	0.00	0.04	0.00	0.23
	10	20	mean	0.56	0.74	0.69	0.76	0.79	0.02	0.08	3.01	0.07	0.69
			std	0.02	0.04	0.06	0.04	0.04	0.00	0.00	0.47	0.07	0.36
0.5	4	10	mean	0.60	0.57	0.39	0.65	0.72	0.01	0.01	0.08	0.02	0.21
			std	0.05	0.06	0.06	0.06	0.07	0.00	0.00	0.01	0.00	0.09
	7	15	mean	0.58	0.58	0.42	0.67	0.80	0.01	0.01	0.18	0.04	0.52
			std	0.03	0.06	0.10	0.05	0.05	0.00	0.00	0.06	0.00	0.29
	10	20	mean	0.56	0.60	0.46	0.69	0.86	0.02	0.08	5.21	0.06	1.30
			std	0.02	0.04	0.07	0.04	0.04	0.00	0.00	1.29	0.00	0.61

Table 2: Performance with additive homogeneous reservation prices uniform

We present the mean and standard deviation of the expected profit of the algorithms relative to the clairvoyant, as well as the mean and standard deviation of their running time in seconds. The summarized results for the uniform distribution can be found in Tables 2 and 3. Furthermore, detailed results for other distribution settings are provided in the Appendix, specifically in Tables A.1 to A.6.

				Performance					Time in Second				
λ	n	m	Descriptive Statistics	R_COM	R_PBDC	R_LST	R_PAH	R_EPAH	T_COM	T_PBDC	T_LST	T_PAH	T_EPAH
0	4	10	mean	0.52	0.56	0.46	0.58	0.60	0.01	0.01	0.07	0.02	0.28
			std	0.05	0.06	0.07	0.06	0.06	0.00	0.00	0.02	0.00	0.07
	7	15	mean	0.49	0.55	0.47	0.59	0.61	0.01	0.01	0.18	0.03	0.97
			std	0.02	0.04	0.04	0.04	0.04	0.00	0.00	0.03	0.00	0.15
	10	20	mean	0.47	0.54	0.47	0.58	0.61	0.02	0.07	2.91	0.05	3.00
			std	0.02	0.04	0.04	0.03	0.04	0.00	0.00	0.46	0.00	0.46
0.5	4	10	mean	0.56	0.55	0.30	0.61	0.66	0.00	0.00	0.07	0.02	0.22
			std	0.06	0.07	0.06	0.07	0.08	0.00	0.00	0.01	0.00	0.07
	7	15	mean	0.52	0.51	0.32	0.59	0.67	0.01	0.01	0.19	0.03	0.88
			std	0.04	0.06	0.03	0.06	0.06	0.00	0.00	0.05	0.00	0.18
	10	20	mean	0.51	0.51	0.32	0.59	0.69	0.02	0.08	5.08	0.04	2.98
			std	0.02	0.05	0.04	0.05	0.05	0.00	0.01	1.30	0.00	0.64

Table 3: Performance with additive heterogeneous reservation prices uniform

The findings demonstrate that the PAH and E-PAH algorithms outperform all other heuristics, and the difference in their performance compared to other heuristics increases with the number of products (n) and the number of consumer types (m). The results indicate that while the performance of other algorithms can significantly deteriorate with increasing costs, our heuristics remain relatively insensitive to changes in prices. In other words, the superiority of our heuristics is more pronounced for $\lambda = 0.5$ and for larger n and m. Specifically, PAH can achieve a performance that is 21% better than the maximum of all previous heuristics, and E-PAH can reach a profit level of up to 56% of the maximum among all previous heuristics.

Furthermore, while the performance of PAH surpasses that of LST, its running time is comparatively lower. The running time of LST, PAH, and E-PAH is sensitive to the number of products and consumer types, with LST exhibiting higher sensitivity. Specifically, the running time of LST can increase up to 100 times that of PAH and 10 times that of E-PAH, depending on the number of products and consumer types.

We also observe that the additional step in E-PAH demonstrates greater efficiency within non-zero cost scenarios. The results illustrate that while E-PAH achieves a profit level of up to 6% relative to PAH in zero-cost situations, this margin significantly improves to 28% in non-zero cost scenarios. Therefore, we suggest that businesses consider implementing E-PAH when operating in contexts involving non-zero costs.

7.2 Symmetric Reservation Prices

To reduce the amount of gathered information and provide a more realistic representation of real-world scenarios, an alternative approach involves considering the inter-relatedness of each pair of products in conjunction with the reservation prices of individual products. This approach, as demonstrated by Ye et al. (2017), effectively captures the relationships between products while maintaining a manageable number of queries, specifically at the order of $O(n^2)$ per consumer type. In this case, information about each consumer type $i \in I$ is summarized in an n-dimensional symmetric matrix $\boldsymbol{A}^i$, where the reservation price for bundle $\boldsymbol{x}$ takes the form $v_i(\boldsymbol{x}) = \boldsymbol{x}^T \boldsymbol{A}^j \boldsymbol{x}$. The off-diagonal elements of $\boldsymbol{A}^i$ indicate whether products are complements or substitutes for consumer type i (positive or negative values) or independent (zero values).

In this context, we propose the utilization of the Shapley value as an efficient starting point for our heuristics. A detailed analysis of the Shapley value, demonstrating its polynomial computation within the symmetric matrices, is provided in Appendix D to ensure the paper's conciseness. To distinguish it from the ones that start with pure bundling, we introduce the abbreviation SH. We denote the algorithms as PAH-SH and EPAH-SH, representing the PAH algorithms and E-PAH algorithms, respectively, with the Shapley value as the initial starting point.

Again, we consider two cases: homogeneous and heterogeneous cases. Specifically, for the homogeneous case, we assume that consumers' reservation prices are provided by a symmetric matrix in which the diagonal represents the reservation prices of consumers for each product. All consumers' reservation prices for single products are derived from the same distribution, while off-diagonal elements are equal to $\beta = \frac{\min\{v_i(\boldsymbol{x}):i\in I,|\boldsymbol{x}|=1\}}{n}$ for all consumer types. This case refers to substitutable products with homogeneous substitutability. Additionally, we assume $\mu_i = \mu = 0.5$ for all $i \in [n]$.

In the heterogeneous case, we assume that that consumers' reservation prices for individual products are drawn from the same distribution, but with distinct expected value denoted by μ_i. Specifically, we set $\mu_i = \mu \frac{i}{m}$, where $\mu = 0.5$. Additionally, we assume there exists varying degrees of inter-relatedness among the products. With a probability of 0.5, we assume that all consumers perceive the products as substitutable, indicating a consistent sign of inter-relatedness. Alternatively, with a probability of 0.5, we allow for randomness in the inter-relatedness, where products can be substitutable for some consumers while being complements for others. This randomness is incorporated to ensure a realistic representation of consumer preferences.

To prevent the consumers' reservation prices from decreasing as the size of the consumer group increases, we consider that the substitutability and complementarity rates are dependent on the individual consumers' reservation prices for each product. Let a^i_{jk} denote the element in column k and row j of matrix $\boldsymbol{A}^i$. Specifically, with a probability of 0.5, we assign a random value to a^i_{jk} extracted from a uniform distribution within the range $[-\min\{a^i_{jj}, a^i_{kk}\}, 0]$. With the remaining probability of 0.5, we extract the value from a uniform distribution within the range $[-\min\{a^i_{jj}, a^i_{kk}\}, \min\{a^i_{jj}, a^i_{kk}\}]$. This stochastic approach introduces variability in the inter-relatedness of products, capturing the diverse nature of consumer preferences.

				Performance							Time in Second						
λ	n	m	Descriptive Statistics	R_COM	R_PBDC	R_LST	R_PAH	R_EPAH	R_PAH_SH	R_EPAH_SH	T_COM	T_PBDC	T_LST	T_PAH	T_EPAH	T_PAH_SH	T_EPAH_SH
0	4	10	mean	0.52	0.68	0.60	0.70	0.72	0.68	0.75	0.01	0.01	0.08	0.03	0.17	0.03	0.31
			std	0.04	0.05	0.06	0.05	0.06	0.07	0.05	0.00	0.00	0.02	0.00	0.09	0.00	0.06
	7	15	mean	0.53	0.73	0.66	0.75	0.77	0.76	0.83	0.01	0.02	0.19	0.04	0.32	0.04	0.88
			std	0.02	0.04	0.07	0.04	0.04	0.05	0.03	0.00	0.00	0.04	0.00	0.18	0.00	0.15
	10	20	mean	0.54	0.74	0.68	0.77	0.79	0.80	0.86	0.02	0.08	3.12	0.08	0.77	0.07	2.36
			std	0.02	0.04	0.08	0.04	0.05	0.04	0.03	0.00	0.01	0.50	0.10	0.45	0.00	0.45
0.5	4	10	mean	0.52	0.56	0.39	0.64	0.72	0.74	0.77	0.01	0.01	0.09	0.02	0.20	0.03	0.24
			std	0.05	0.06	0.05	0.07	0.06	0.08	0.06	0.00	0.00	0.01	0.00	0.08	0.00	0.07
	7	15	mean	0.53	0.57	0.42	0.66	0.80	0.82	0.86	0.01	0.01	0.19	0.04	0.55	0.04	0.91
			std	0.03	0.06	0.06	0.05	0.06	0.05	0.04	0.00	0.00	0.06	0.00	0.28	0.00	0.17
	10	20	mean	0.54	0.59	0.46	0.68	0.86	0.87	0.91	0.02	0.08	5.19	0.06	1.30	0.07	2.51
			std	0.03	0.04	0.04	0.04	0.04	0.06	0.03	0.00	0.00	1.33	0.00	0.58	0.00	0.56

Table 4: Performance with symmetric homogeneous reservation prices uniform

Tables 4 and 5, along with Tables A.7-A.12 in Appendix A, illustrate the results for both homogeneous and heterogeneous cases in the symmetric scenario. We observe that our heuristics, whether initiated from pure bundling or the Shapley value, consistently outperform previous approaches in terms of performance. Additionally, we note that starting from the Shapley value as an initial point leads to higher profits.

7.3 General Reservation Prices

In this subsection, we explore a general condition where comprehensive information regarding all combinations of products is available. We assume that consumers have heterogeneous reservation prices for both individual products and bundles of products, with reservation prices slightly increasing as the

				Performance							Time in Second						
λ	n	m	Descriptive Statistics	R_COM	R_PBDC	R_LST	R_PAH	R_EPAH	R_PAH_SH	R_EPAH_SH	T_COM	T_PBDC	T_LST	T_PAH	T_EPAH	T_PAH_SH	T_EPAH_SH
0	4	10	mean	0.43	0.56	0.45	0.58	0.61	0.58	0.64	0.01	0.01	0.07	0.02	0.27	0.02	0.33
			std	0.05	0.05	0.06	0.06	0.06	0.08	0.06	0.00	0.00	0.01	0.00	0.06	0.00	0.06
	7	15	mean	0.40	0.53	0.46	0.57	0.61	0.60	0.67	0.01	0.01	0.18	0.03	1.01	0.03	1.22
			std	0.03	0.04	0.06	0.05	0.05	0.05	0.03	0.00	0.00	0.03	0.00	0.18	0.00	0.15
	10	20	mean	0.38	0.53	0.45	0.58	0.62	0.63	0.69	0.02	0.08	3.14	0.05	3.18	0.05	3.89
			std	0.02	0.04	0.07	0.04	0.05	0.06	0.03	0.00	0.00	0.51	0.01	0.56	0.00	0.45
0.5	4	10	mean	0.46	0.55	0.36	0.60	0.66	0.65	0.69	0.00	0.00	0.08	0.02	0.25	0.02	0.26
			std	0.06	0.07	0.05	0.07	0.07	0.10	0.07	0.00	0.00	0.01	0.00	0.07	0.00	0.05
	7	15	mean	0.41	0.53	0.39	0.59	0.66	0.68	0.74	0.01	0.01	0.20	0.03	0.96	0.03	1.11
			std	0.04	0.06	0.06	0.06	0.07	0.07	0.05	0.00	0.00	0.05	0.00	0.24	0.00	0.19
	10	20	mean	0.38	0.52	0.40	0.58	0.67	0.69	0.75	0.02	0.07	5.17	0.05	3.04	0.05	3.69
			std	0.03	0.05	0.07	0.06	0.06	0.06	0.03	0.00	0.00	1.34	0.00	0.85	0.00	0.57

Table 5: Performance with symmetric heterogeneous reservation prices uniform

bundle size grows. To meet these conditions, we extract consumers' reservation prices for individual products from distributions with a mean $\mu_i = \mu \frac{i}{m}$, where $\mu = 0.5$. Additionally, we assume substitutability among the products and derive the substitutability rate for any combination of products from a uniform distribution, ensuring that reservation prices remain non-decreasing with bundle sizes.

Furthermore, we consider that products have non-additive heterogeneous costs, resulting in lower costs as bundle sizes increase. This assumption aligns with real-world scenarios, as selling products in a bundle often translates to reduced marketing and selling expenses (Fox, 2014).

			Performance				Time in Second			
n	m	Descriptive Statistics	R_COM	R_PBDC	R_PAH	R_EPAH	T_COM	T_PBDC	T_PAH	T_EPAH
4	10	mean	0.04	0.05	0.07	0.08	0.00	0.01	0.02	0.17
		std	0.01	0.01	0.01	0.01	0.00	0.00	0.00	0.07
7	15	mean	0.02	0.04	0.05	0.05	0.01	0.01	0.03	0.46
		std	0.00	0.01	0.00	0.00	0.00	0.00	0.00	0.19
10	20	mean	0.02	0.03	0.03	0.04	0.02	0.08	0.06	1.07
		std	0.00	0.00	0.00	0.00	0.00	0.01	0.01	0.59

Table 6: Performance with general reservation prices uniform

Table 6 and tables A.13-A.15 in Appendix A provide evidence supporting our expectation that the performance of all heuristics would deteriorate for general cases. This outcome is expected as no heuristic guarantees a better worst-case performance than zero for bundle pricing (Hart, Nisan, et al., 2013). Nevertheless, it is important to acknowledge that despite the decline in performance, our heuristics still exhibit significant improvements compared to alternative approaches. Specifically, the E-PAH heuristic generated profits at least 33% higher than the maximum profit achieved by PBDC and COM,with some cases showing an increase of up to 64% compared to them.

8 Conclusion

In this study, we have introduced a first general heuristic approach for bundle pricing problem, addressing the fundamental structure of the problem by breaking it down into pricing and assortment optimization subproblems. Our proposed heuristic, named PAH, has demonstrated superior performance compared to existing approaches in both numerical and analytical solutions. Even in the presence of costs and consumer heterogeneity, the PAH algorithm has been effective in offering a wide range of products, leading to improved profits and social efficiency by catering to a diverse consumer types.

The flexibility of our heuristic allows it to start from any feasible point. However, this flexibility also introduces a potential drawback, as the final solution may be dependent on the initial point and could get trapped in local optima. To overcome this limitation, we have presented an extended version of the heuristic called E-PAH. This algorithm incorporates an additional step wherein a slight deviation from the obtained solution is made, thereby reducing sensitivity to the starting point and further improving the objective function.

The success of our algorithms in bundle pricing opens up intriguing possibilities for exploring alternative applications. Future research can focus on modifying the algorithm to accommodate a random utility choice model, further expanding its versatility. Additionally, investigating the efficiency of the algorithm for dynamic bundling problems presents an interesting avenue for further study.

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Appendices

A Robustness of Numerical Study

In this section, we demonstrate the robustness of our algorithms through numerical studies conducted on three distinct distributions: "exponential," "lognormal," and "truncated normal". By examining these varied scenarios, we aim to provide comprehensive evidence of the effectiveness and reliability of our proposed approaches.

				Performance					Time in Second				
λ	n	m	Descriptive Statistics	R_COM	R_PBDC	R_LST	R_PAH	R_EPAH	T_COM	T_PBDC	T_LST	T_PAH	T_EPAH
0	4	10	mean	0.50	0.59	0.50	0.60	0.63	0.01	0.01	0.08	0.02	0.22
			std	0.06	0.06	0.09	0.06	0.06	0.00	0.00	0.01	0.00	0.09
	7	15	mean	0.46	0.62	0.54	0.64	0.67	0.01	0.01	0.18	0.04	0.56
			std	0.03	0.05	0.07	0.05	0.06	0.00	0.00	0.04	0.01	0.24
	10	20	mean	0.44	0.64	0.58	0.67	0.70	0.02	0.08	3.92	0.06	1.26
			std	0.02	0.05	0.06	0.05	0.05	0.00	0.01	0.96	0.01	0.65
0.5	4	10	mean	0.55	0.50	0.30	0.58	0.66	0.01	0.01	0.08	0.02	0.26
			std	0.07	0.07	0.05	0.08	0.07	0.00	0.00	0.02	0.00	0.08
	7	15	mean	0.50	0.48	0.34	0.59	0.72	0.01	0.01	0.18	0.03	0.93
			std	0.05	0.06	0.08	0.06	0.06	0.00	0.00	0.06	0.00	0.29
	10	20	mean	0.47	0.50	0.35	0.60	0.77	0.02	0.08	5.36	0.06	2.57
			std	0.03	0.05	0.09	0.06	0.05	0.00	0.01	2.24	0.01	1.10

Table A.1: Performance with additive homogeneous reservation prices exponential

				Performance					Time in Second				
λ	n	m	Descriptive Statistics	R_COM	R_PBDC	R_LST	R_PAH	R_EPAH	T_COM	T_PBDC	T_LST	T_PAH	T_EPAH
0	4	10	mean	0.46	0.60	0.49	0.60	0.62	0.01	0.01	0.07	0.03	0.18
			std	0.07	0.08	0.12	0.08	0.07	0.00	0.00	0.02	0.00	0.09
	7	15	mean	0.43	0.64	0.55	0.64	0.66	0.01	0.01	0.18	0.04	0.40
			std	0.04	0.05	0.09	0.06	0.05	0.00	0.00	0.06	0.00	0.25
	10	20	mean	0.41	0.65	0.55	0.66	0.68	0.02	0.08	4.89	0.07	1.09
			std	0.03	0.04	0.13	0.06	0.05	0.00	0.00	1.25	0.01	0.50
0.5	4	10	mean	0.53	0.51	0.34	0.56	0.60	0.01	0.01	0.08	0.02	0.28
			std	0.09	0.07	0.07	0.07	0.07	0.00	0.00	0.02	0.00	0.08
	7	15	mean	0.47	0.49	0.38	0.55	0.62	0.01	0.01	0.18	0.03	0.98
			std	0.07	0.06	0.07	0.07	0.06	0.00	0.00	0.07	0.00	0.32
	10	20	mean	0.45	0.50	0.41	0.57	0.67	0.02	0.08	5.91	0.06	2.81
			std	0.04	0.05	0.09	0.06	0.05	0.00	0.00	1.81	0.01	1.18

Table A.2: Performance with additive homogeneous reservation prices lognormal

B Impact of Escaping from Initial Point

In this section, we will conduct a numerical study to illustrate the impact of implementing the "escaping from initial point" step in the E-PAH algorithm. Our objective is to demonstrate that E-PAH consistently achieves a significantly higher expected profit compared to PAH, while also exhibiting a lower standard

				Performance					Time in Second				
λ	n	m	Descriptive Statistics	R_COM	R_PBDC	R_LST	R_PAH	R_EPAH	T_COM	T_PBDC	T_LST	T_PAH	T_EPAH
0	4	10	mean	0.56	0.72	0.62	0.73	0.73	0.01	0.01	0.07	0.03	0.11
			std	0.04	0.05	0.12	0.05	0.05	0.00	0.00	0.02	0.00	0.08
	7	15	mean	0.54	0.76	0.68	0.77	0.77	0.01	0.02	0.17	0.04	0.21
			std	0.02	0.04	0.11	0.03	0.03	0.00	0.00	0.05	0.00	0.18
	10	20	mean	0.53	0.78	0.70	0.79	0.80	0.02	0.08	4.58	0.07	0.46
			std	0.02	0.03	0.13	0.03	0.03	0.00	0.01	0.93	0.01	0.42
0.5	4	10	mean	0.54	0.61	0.44	0.65	0.67	0.01	0.01	0.07	0.03	0.23
			std	0.05	0.06	0.09	0.06	0.06	0.00	0.00	0.02	0.00	0.09
	7	15	mean	0.51	0.63	0.48	0.67	0.72	0.01	0.01	0.18	0.04	0.55
			std	0.03	0.05	0.14	0.05	0.06	0.00	0.00	0.08	0.00	0.24
	10	20	mean	0.49	0.65	0.55	0.69	0.75	0.02	0.08	5.84	0.06	1.40
			std	0.02	0.04	0.07	0.04	0.05	0.00	0.01	2.84	0.00	0.92

Table A.3: Performance with additive homogeneous reservation prices truncated normal

				Performance					Time in Second				
λ	n	m	Descriptive Statistics	R_COM	R_PBDC	R_LST	R_PAH	R_EPAH	T_COM	T_PBDC	T_LST	T_PAH	T_EPAH
0	4	10	mean	0.48	0.54	0.43	0.56	0.58	0.01	0.01	0.07	0.02	0.28
			std	0.09	0.07	0.09	0.08	0.07	0.00	0.00	0.01	0.00	0.07
	7	15	mean	0.44	0.52	0.44	0.56	0.59	0.01	0.01	0.18	0.03	0.99
			std	0.04	0.05	0.06	0.05	0.05	0.00	0.00	0.03	0.00	0.20
	10	20	mean	0.43	0.52	0.45	0.56	0.59	0.02	0.07	3.84	0.05	3.05
			std	0.03	0.04	0.07	0.05	0.05	0.00	0.00	0.87	0.00	0.52
0.5	4	10	mean	0.56	0.52	0.28	0.59	0.66	0.00	0.00	0.07	0.02	0.23
			std	0.10	0.07	0.06	0.07	0.09	0.00	0.00	0.01	0.00	0.06
	7	15	mean	0.50	0.48	0.28	0.57	0.67	0.01	0.01	0.18	0.03	0.89
			std	0.05	0.06	0.05	0.07	0.07	0.00	0.00	0.05	0.00	0.22
	10	20	mean	0.48	0.47	0.29	0.56	0.69	0.02	0.07	5.21	0.05	2.92
			std	0.04	0.05	0.06	0.06	0.05	0.00	0.01	1.92	0.00	0.77

Table A.4: Performance with additive heterogeneous reservation prices exponential

				Performance					Time in Second				
λ	n	m	Descriptive Statistics	R_COM	R_PBDC	R_LST	R_PAH	R_EPAH	T_COM	T_PBDC	T_LST	T_PAH	T_EPAH
0	4	10	mean	0.46	0.58	0.45	0.59	0.60	0.01	0.01	0.08	0.02	0.19
			std	0.08	0.08	0.13	0.09	0.07	0.00	0.00	0.02	0.00	0.10
	7	15	mean	0.43	0.60	0.51	0.61	0.63	0.01	0.01	0.18	0.04	0.46
			std	0.04	0.05	0.08	0.06	0.06	0.00	0.00	0.05	0.00	0.25
	10	20	mean	0.41	0.63	0.54	0.63	0.66	0.02	0.08	4.85	0.06	1.26
			std	0.03	0.05	0.09	0.05	0.05	0.00	0.00	1.27	0.01	0.77
0.5	4	10	mean	0.56	0.51	0.31	0.56	0.63	0.01	0.00	0.08	0.02	0.26
			std	0.10	0.07	0.08	0.08	0.08	0.00	0.00	0.02	0.00	0.08
	7	15	mean	0.50	0.47	0.35	0.55	0.64	0.01	0.01	0.18	0.03	0.99
			std	0.07	0.05	0.07	0.06	0.07	0.00	0.00	0.07	0.00	0.31
	10	20	mean	0.45	0.48	0.38	0.56	0.68	0.02	0.08	5.66	0.05	3.09
			std	0.05	0.05	0.08	0.05	0.05	0.00	0.00	1.77	0.01	1.52

Table A.5: Performance with additive heterogeneous reservation prices lognormal

deviation. This lower standard deviation indicates reduced dependence on the initial point, highlighting the robustness of the E-PAH algorithm.

We consider that consumers have additive homogeneous reservation prices. A detailed explanation of the consumers' reservation prices can be found in Section 7.1. We consider eight different starting

				Performance					Time in Second				
λ	n	m	Descriptive Statistics	R_COM	R_PBDC	R_LST	R_PAH	R_EPAH	T_COM	T_PBDC	T_LST	T_PAH	T_EPAH
0	4	10	mean	0.54	0.68	0.57	0.69	0.70	0.01	0.01	0.07	0.02	0.15
			std	0.04	0.06	0.14	0.06	0.05	0.00	0.00	0.02	0.00	0.09
	7	15	mean	0.51	0.69	0.62	0.71	0.73	0.01	0.01	0.17	0.04	0.40
			std	0.02	0.05	0.09	0.04	0.04	0.00	0.00	0.06	0.00	0.23
	10	20	mean	0.51	0.69	0.61	0.71	0.75	0.02	0.08	4.42	0.06	0.94
			std	0.02	0.04	0.12	0.05	0.05	0.00	0.00	0.88	0.00	0.53
0.5	4	10	mean	0.54	0.57	0.39	0.63	0.67	0.01	0.01	0.07	0.02	0.24
			std	0.05	0.06	0.07	0.07	0.06	0.00	0.00	0.02	0.00	0.07
	7	15	mean	0.51	0.56	0.43	0.63	0.71	0.01	0.01	0.17	0.03	0.66
			std	0.03	0.05	0.06	0.05	0.06	0.00	0.00	0.06	0.00	0.28
	10	20	mean	0.49	0.57	0.46	0.65	0.75	0.02	0.08	5.39	0.06	1.99
			std	0.02	0.04	0.07	0.04	0.05	0.00	0.00	3.46	0.01	1.21

Table A.6: Performance with additive heterogeneous reservation prices truncated normal

				Performance							Time in Second						
λ	n	m	Descriptive Statistics	R_COM	R_PBDC	R_LST	R_PAH	R_EPAH	R_PAH_SH	R_EPAH_SH	T_COM	T_PBDC	T_LST	T_PAH	T_EPAH	T_PAH_SH	T_EPAH_SH
0	4	10	mean	0.42	0.59	0.50	0.60	0.63	0.59	0.65	0.01	0.01	0.08	0.02	0.22	0.03	0.32
			std	0.06	0.06	0.07	0.06	0.06	0.07	0.07	0.00	0.00	0.02	0.00	0.10	0.00	0.06
	7	15	mean	0.42	0.62	0.54	0.64	0.66	0.66	0.72	0.01	0.01	0.19	0.04	0.57	0.04	1.01
			std	0.04	0.05	0.08	0.06	0.05	0.06	0.05	0.00	0.00	0.03	0.01	0.24	0.00	0.22
	10	20	mean	0.43	0.64	0.57	0.67	0.70	0.72	0.77	0.02	0.08	4.07	0.07	1.30	0.07	2.88
			std	0.03	0.04	0.10	0.05	0.05	0.06	0.04	0.00	0.00	0.54	0.01	0.65	0.00	0.65
0.5	4	10	mean	0.49	0.48	0.30	0.56	0.66	0.63	0.66	0.01	0.01	0.08	0.02	0.26	0.03	0.25
			std	0.09	0.07	0.07	0.08	0.08	0.10	0.08	0.00	0.00	0.02	0.00	0.08	0.00	0.06
	7	15	mean	0.45	0.48	0.33	0.58	0.71	0.68	0.73	0.01	0.01	0.19	0.03	0.92	0.04	1.04
			std	0.06	0.06	0.08	0.06	0.06	0.07	0.06	0.00	0.00	0.06	0.00	0.34	0.00	0.28
	10	20	mean	0.45	0.50	0.36	0.61	0.77	0.76	0.80	0.02	0.08	5.54	0.06	2.27	0.07	2.98
			std	0.04	0.05	0.09	0.05	0.05	0.06	0.05	0.00	0.00	1.18	0.01	0.83	0.00	0.80

Table A.7: Performance with symmetric homogeneous reservation prices exponential

				Performance							Time in Second						
λ	n	m	Descriptive Statistics	R_COM	R_PBDC	R_LST	R_PAH	R_EPAH	R_PAH_SH	R_EPAH_SH	T_COM	T_PBDC	T_LST	T_PAH	T_EPAH	T_PAH_SH	T_EPAH_SH
0	4	10	mean	0.36	0.58	0.46	0.58	0.60	0.56	0.60	0.01	0.01	0.08	0.02	0.20	0.03	0.29
			std	0.08	0.07	0.12	0.08	0.07	0.07	0.07	0.00	0.00	0.02	0.00	0.10	0.00	0.06
	7	15	mean	0.34	0.60	0.51	0.62	0.63	0.60	0.65	0.01	0.01	0.19	0.04	0.49	0.04	0.93
			std	0.05	0.06	0.10	0.06	0.06	0.06	0.06	0.00	0.00	0.07	0.00	0.27	0.00	0.20
	10	20	mean	0.34	0.63	0.55	0.65	0.67	0.66	0.70	0.02	0.08	5.05	0.07	1.27	0.07	2.57
			std	0.03	0.05	0.07	0.05	0.04	0.06	0.05	0.00	0.01	1.26	0.01	0.63	0.00	0.66
0.5	4	10	mean	0.45	0.48	0.25	0.57	0.64	0.54	0.58	0.01	0.01	0.08	0.02	0.28	0.03	0.21
			std	0.13	0.07	0.07	0.07	0.08	0.11	0.09	0.00	0.00	0.02	0.00	0.07	0.00	0.06
	7	15	mean	0.38	0.45	0.31	0.55	0.67	0.60	0.64	0.01	0.01	0.19	0.03	1.00	0.04	0.83
			std	0.08	0.07	0.07	0.07	0.08	0.09	0.08	0.00	0.00	0.06	0.00	0.32	0.00	0.25
	10	20	mean	0.34	0.44	0.32	0.56	0.71	0.64	0.68	0.02	0.08	4.87	0.05	3.16	0.07	2.60
			std	0.06	0.06	0.08	0.06	0.06	0.07	0.06	0.00	0.00	2.54	0.01	1.20	0.00	0.89

Table A.8: Performance with symmetric homogeneous reservation prices lognormal

points: COM, pure bundling, PBDC, and five random starting points. For each combination of n, m, λ, and distribution, we calculate both the expected profit and the standard deviation of the expected profit obtained by employing the PAH and E-PAH algorithms from these eight initial points. To ensure statistical robustness, we perform 100 replicates for each combination.

To quantify the improvement achieved by the E-PAH algorithm, we analyze the impact of escaping from the initial point and report the percentage increase in expected profit on average. This is calculated

λ	n	m	Descriptive Statistics	Performance							Time in Second						
				R_COM	R_PBDC	R_LST	R_PAH	R_EPAH	R_PAH_SH	R_EPAH_SH	T_COM	T_PBDC	T_LST	T_PAH	T_EPAH	T_PAH_SH	T_EPAH_SH
0	4	10	mean	0.43	0.70	0.59	0.71	0.72	0.66	0.71	0.01	0.01	0.08	0.03	0.12	0.03	0.26
			std	0.04	0.05	0.13	0.06	0.05	0.06	0.06	0.00	0.00	0.03	0.00	0.07	0.00	0.06
	7	15	mean	0.43	0.73	0.64	0.74	0.75	0.74	0.78	0.01	0.02	0.18	0.04	0.26	0.04	0.75
			std	0.02	0.05	0.11	0.04	0.04	0.05	0.04	0.00	0.00	0.05	0.00	0.18	0.00	0.17
	10	20	mean	0.43	0.74	0.67	0.76	0.78	0.80	0.82	0.02	0.08	4.59	0.07	0.65	0.07	1.95
			std	0.02	0.04	0.12	0.04	0.04	0.03	0.03	0.00	0.01	0.96	0.00	0.47	0.00	0.37
0.5	4	10	mean	0.45	0.52	0.32	0.61	0.69	0.63	0.66	0.01	0.01	0.08	0.02	0.24	0.03	0.18
			std	0.06	0.06	0.11	0.06	0.07	0.09	0.08	0.00	0.00	0.03	0.00	0.09	0.00	0.06
	7	15	mean	0.42	0.53	0.34	0.64	0.78	0.74	0.76	0.01	0.01	0.18	0.04	0.71	0.04	0.52
			std	0.03	0.06	0.10	0.05	0.05	0.07	0.06	0.00	0.00	0.08	0.00	0.29	0.00	0.16
	10	20	mean	0.42	0.55	0.39	0.67	0.84	0.82	0.83	0.02	0.08	5.01	0.06	1.83	0.07	1.37
			std	0.02	0.04	0.10	0.04	0.04	0.05	0.05	0.00	0.00	2.48	0.01	0.97	0.00	0.42

Table A.9: Performance with symmetric homogeneous reservation prices truncated normal

λ	n	m	Descriptive Statistics	Performance							Time in Second						
				R_COM	R_PBDC	R_LST	R_PAH	R_EPAH	R_PAH_SH	R_EPAH_SH	T_COM	T_PBDC	T_LST	T_PAH	T_EPAH	T_PAH_SH	T_EPAH_SH
0	4	10	mean	0.42	0.54	0.41	0.57	0.59	0.59	0.64	0.01	0.01	0.07	0.02	0.27	0.02	0.33
			std	0.06	0.06	0.09	0.06	0.06	0.09	0.06	0.00	0.00	0.01	0.00	0.07	0.00	0.05
	7	15	mean	0.37	0.52	0.43	0.56	0.59	0.62	0.67	0.01	0.01	0.19	0.03	1.04	0.03	1.24
			std	0.04	0.05	0.07	0.05	0.05	0.06	0.04	0.00	0.00	0.03	0.00	0.25	0.00	0.17
	10	20	mean	0.36	0.52	0.44	0.56	0.61	0.65	0.70	0.02	0.08	4.05	0.05	3.29	0.05	4.03
			std	0.03	0.05	0.07	0.05	0.06	0.05	0.03	0.00	0.00	0.58	0.00	0.66	0.03	0.46
0.5	4	10	mean	0.48	0.52	0.32	0.58	0.66	0.66	0.69	0.00	0.00	0.07	0.02	0.23	0.02	0.24
			std	0.08	0.07	0.06	0.08	0.07	0.11	0.09	0.00	0.00	0.01	0.00	0.07	0.00	0.07
	7	15	mean	0.40	0.49	0.35	0.56	0.66	0.69	0.73	0.01	0.01	0.20	0.03	1.06	0.03	1.22
			std	0.05	0.06	0.05	0.07	0.07	0.07	0.05	0.00	0.00	0.07	0.00	0.28	0.00	0.21
	10	20	mean	0.37	0.50	0.39	0.57	0.68	0.72	0.77	0.02	0.07	5.43	0.05	3.23	0.05	3.84
			std	0.04	0.05	0.04	0.05	0.06	0.06	0.04	0.00	0.01	1.21	0.00	0.89	0.01	0.61

Table A.10: Performance with symmetric heterogeneous reservation prices exponential

λ	n	m	Descriptive Statistics	Performance							Time in Second						
				R_COM	R_PBDC	R_LST	R_PAH	R_EPAH	R_PAH_SH	R_EPAH_SH	T_COM	T_PBDC	T_LST	T_PAH	T_EPAH	T_PAH_SH	T_EPAH_SH
0	4	10	mean	0.41	0.59	0.49	0.59	0.61	0.53	0.63	0.01	0.01	0.08	0.02	0.18	0.02	0.34
			std	0.06	0.07	0.11	0.08	0.07	0.08	0.07	0.00	0.00	0.02	0.00	0.08	0.00	0.08
	7	15	mean	0.38	0.61	0.54	0.62	0.66	0.54	0.68	0.01	0.02	0.18	0.04	0.50	0.04	1.14
			std	0.05	0.05	0.08	0.07	0.05	0.07	0.05	0.00	0.00	0.05	0.00	0.23	0.01	0.21
	10	20	mean	0.38	0.63	0.55	0.65	0.69	0.57	0.73	0.02	0.08	5.07	0.06	1.22	0.06	2.73
			std	0.04	0.05	0.08	0.05	0.05	0.06	0.04	0.00	0.00	1.29	0.01	0.69	0.01	0.70
0.5	4	10	mean	0.46	0.51	0.34	0.56	0.60	0.59	0.65	0.01	0.01	0.07	0.02	0.27	0.02	0.33
			std	0.08	0.07	0.10	0.07	0.07	0.10	0.08	0.00	0.00	0.02	0.00	0.08	0.00	0.07
	7	15	mean	0.37	0.51	0.43	0.56	0.63	0.61	0.70	0.01	0.01	0.20	0.04	0.88	0.04	1.36
			std	0.06	0.05	0.07	0.06	0.07	0.07	0.06	0.00	0.00	0.09	0.00	0.29	0.01	0.26
	10	20	mean	0.34	0.54	0.47	0.58	0.66	0.61	0.73	0.02	0.08	5.20	0.06	2.04	0.06	3.95
			std	0.05	0.05	0.10	0.05	0.05	0.06	0.04	0.00	0.00	2.44	0.01	0.98	0.01	1.34

Table A.11: Performance with symmetric heterogeneous reservation prices lognormal

as $\frac{E[R_{EPAH}]-E[R_{PAH}]}{E[R_{PAH}]}$. Additionally, we assess the percentage decrease in the standard deviation of the expected profit resulting from escaping from the initial point, represented by $\frac{\sigma(R_{PAH})-\sigma(R_{EPAH})}{\sigma(R_{PAH})}$. These metrics offer a comprehensive understanding of the benefits of the escaping from initial point step, highlighting improvements in both expected profit and reduced dependence on the initial point.

Table B.1 highlights the significant improvements achieved through the added step in the E-PAH algorithm, which allows for escaping local maxima. On average, "escaping from initial point" step leads to a noteworthy increase in expected profit of at least 32%, with the potential for even greater

				Performance							Time in Second						
λ	n	m	Descriptive Statistics	R_COM	R_PBDC	R_LST	R_PAH	R_EPAH	R_PAH_SH	R_EPAH_SH	T_COM	T_PBDC	T_LST	T_PAH	T_EPAH	T_PAH_SH	T_EPAH_SH
0	4	10	mean	0.48	0.65	0.56	0.67	0.68	0.54	0.69	0.01	0.01	0.07	0.02	0.15	0.02	0.34
			std	0.05	0.07	0.10	0.07	0.06	0.09	0.07	0.00	0.00	0.02	0.00	0.08	0.00	0.07
	7	15	mean	0.45	0.65	0.59	0.68	0.71	0.56	0.75	0.01	0.01	0.19	0.04	0.41	0.03	1.13
			std	0.03	0.05	0.06	0.05	0.05	0.08	0.04	0.00	0.00	0.05	0.00	0.21	0.00	0.22
	10	20	mean	0.41	0.65	0.59	0.68	0.72	0.55	0.78	0.02	0.08	4.47	0.06	1.08	0.05	2.89
			std	0.02	0.04	0.09	0.04	0.04	0.08	0.03	0.00	0.01	0.98	0.01	0.57	0.01	0.64
0.5	4	10	mean	0.47	0.56	0.44	0.61	0.68	0.60	0.72	0.01	0.01	0.07	0.02	0.22	0.02	0.33
			std	0.06	0.06	0.08	0.06	0.07	0.12	0.06	0.00	0.00	0.02	0.00	0.10	0.00	0.09
	7	15	mean	0.44	0.57	0.50	0.62	0.70	0.61	0.77	0.01	0.01	0.19	0.04	0.68	0.04	1.28
			std	0.03	0.05	0.05	0.05	0.06	0.09	0.04	0.00	0.00	0.07	0.00	0.26	0.01	0.21
	10	20	mean	0.41	0.59	0.52	0.64	0.72	0.58	0.80	0.02	0.08	4.58	0.06	1.46	0.05	3.31
			std	0.03	0.04	0.08	0.04	0.05	0.08	0.03	0.00	0.00	2.08	0.01	0.79	0.01	0.88

Table A.12: Performance with symmetric heterogeneous reservation prices truncated normal

			Performance				Time in Second			
n	m	Descriptive Statistics	R_COM	R_PBDC	R_PAH	R_EPAH	T_COM	T_PBDC	T_PAH	T_EPAH
4	10	mean	0.04	0.05	0.06	0.07	0.00	0.00	0.02	0.23
		std	0.01	0.01	0.01	0.01	0.00	0.00	0.00	0.06
7	15	mean	0.02	0.03	0.04	0.05	0.01	0.01	0.03	0.81
		std	0.00	0.01	0.01	0.01	0.00	0.00	0.00	0.27
10	20	mean	0.01	0.02	0.03	0.04	0.02	0.08	0.05	2.10
		std	0.00	0.00	0.00	0.00	0.00	0.01	0.01	0.73

Table A.13: Performance with general reservation prices exponential

			Performance				Time in Second			
n	m	Descriptive Statistics	R_COM	R_PBDC	R_PAH	R_EPAH	T_COM	T_PBDC	T_PAH	T_EPAH
4	10	mean	0.04	0.05	0.06	0.06	0.00	0.00	0.02	0.25
		std	0.01	0.01	0.01	0.01	0.00	0.00	0.00	0.08
7	15	mean	0.02	0.03	0.04	0.04	0.01	0.01	0.03	0.86
		std	0.01	0.01	0.01	0.01	0.00	0.00	0.00	0.30
10	20	mean	0.01	0.02	0.03	0.04	0.02	0.08	0.05	2.55
		std	0.00	0.00	0.00	0.00	0.00	0.01	0.01	0.95

Table A.14: Performance with general reservation prices log normal

enhancements of up to 50%. Furthermore, the impact of the initial point selection is greatly reduced, as evidenced by the substantial decrease in the standard deviation of the expected profit obtained from different initial points. This reduction ranges from a minimum of 33% to an impressive maximum of 78%.

n	m	Descriptive Statistics	Performance				Time in Second			
			R_COM	R_PBDC	R_PAH	R_EPAH	T_COM	T_PBDC	T_PAH	T_EPAH
4	10	mean	0.04	0.05	0.06	0.07	0.00	0.01	0.02	0.19
		std	0.01	0.01	0.01	0.01	0.00	0.00	0.00	0.08
7	15	mean	0.02	0.04	0.05	0.05	0.01	0.01	0.03	0.51
		std	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.22
10	20	mean	0.01	0.03	0.03	0.04	0.02	0.08	0.06	1.13
		std	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.62

Table A.15: Performance with general reservation prices truncated normal

λ	n	m	Mean_Ratio	Std_Ratio
0	4	10	37%	55%
	7	15	37%	70%
	10	20	33%	75%
0.5	4	10	43%	51%
	7	15	48%	73%
	10	20	45%	78%

Uniform

λ	n	m	Mean_Ratio	Std_Ratio
0	4	10	32%	39%
	7	15	39%	61%
	10	20	40%	71%
0.5	4	10	34%	33%
	7	15	47%	62%
	10	20	50%	74%

Exponential

λ	n	m	Mean_Ratio	Std_Ratio
0	4	10	37%	55%
	7	15	37%	70%
	10	20	33%	75%
0.5	4	10	43%	51%
	7	15	48%	73%
	10	20	45%	78%

Log normal

λ	n	m	Mean_Ratio	Std_Ratio
0	4	10	32%	39%
	7	15	39%	61%
	10	20	40%	71%
0.5	4	10	34%	33%
	7	15	47%	62%
	10	20	50%	74%

Truncated normal

Table B.1: Impact of escaping from initial point step,

C Proofs

Proof Proposition 1: The PAH algorithm begins with a feasible point for the (MB) problem. As the (LP) problem is a special case of the (MB) problem, it is also feasible. The first and second constraints in (LP) represent the IC and IR constraints in the (MB) for consumers who purchase a bundle. Therefore, the primal is feasible, and demonstrating that the dual is feasible is sufficient to arrive at a bounded solution.

Let $u_{ij} = v_i(\boldsymbol{x}(i)) - v_i(\boldsymbol{x}(j))$, where $\boldsymbol{x}(i)$ indicates $y_i(\boldsymbol{x}(i)) = 1$, i.e., the bundle associated with consumer type i. The dual of (LP) would be as follows:

$$
\begin{aligned}
&\min \sum_{\substack{i,j\in\hat{I}\\ i\neq j}} z_{ij}u_{ij} + \sum_{i\in\hat{I}} z_i v_i(\boldsymbol{x}(i))\\
&St. \sum_{j\in\hat{I}} z_{ij} - \sum_{j\in\hat{I}} z_{ji} + z_i \geq \alpha_i \quad \forall i \in \hat{I}\\
&z_{ij}, z_i \geq 0
\end{aligned}
$$

Where z_{ij}s are dual variables associated with IC constraints, and z_is are dual variables associated with IR constraints. It is easy to see that $z_i = \alpha_i$ and $z_{ij} = 0$ is a feasible solution to the dual problem. As a consequence, both primal and dual problems are feasible, and the (LP) problem has a bounded solution. Let us now demonstrate the uniqueness of a bounded solution through contradiction. Assume that the solution is not unique and that $p_1(\boldsymbol{x})$ and $p_2(\boldsymbol{x})$, $\forall \boldsymbol{x} \in \{0,1\}^n$, are the two optimal solutions for (LP). We will show that $p(\boldsymbol{x}) = \max\{p_2(\boldsymbol{x}), p_1(\boldsymbol{x})\}$, $\forall \boldsymbol{x}$, is a feasible solution. It is easy to see that $p(\boldsymbol{x})$ satisfies the second constraint (IR), since

$$
\begin{aligned}
&v_i(\boldsymbol{x}(i)) \geq p_1(\boldsymbol{x}(i))\\
&v_i(\boldsymbol{x}(i)) \geq p_2(\boldsymbol{x}(i))\\
&v_i(\boldsymbol{x}(i)) \geq \max\{p_1(\boldsymbol{x}(i)), p_2(\boldsymbol{x}(i))\}
\end{aligned}
$$

The IC constraint can be rewritten as follows:

$$
\begin{aligned}
&p_1(\boldsymbol{x}(i)) \leq v_i(\boldsymbol{x}(i)) - v_i(\boldsymbol{x}(j)) + p_1(\boldsymbol{x}(j))\\
&\Rightarrow p_1(\boldsymbol{x}(i)) \leq v_i(\boldsymbol{x}(i)) - v_i(\boldsymbol{x}(j)) + \max\{p_1(\boldsymbol{x}(j)), p_2(\boldsymbol{x}(j))\}
\end{aligned}
$$

Similarly,

$$
\begin{aligned}
&p_2(\boldsymbol{x}(i)) \leq v_i(\boldsymbol{x}(i)) - v_i(\boldsymbol{x}(j)) + \max\{p_1(\boldsymbol{x}(j)), p_2(\boldsymbol{x}(j))\}\\
&\Rightarrow \max\{p_1(\boldsymbol{x}(i)), p_2(\boldsymbol{x}(i))\} \leq v_i(\boldsymbol{x}(i)) - v_i(\boldsymbol{x}(j)) + \max\{p_1(\boldsymbol{x}(j)), p_2(\boldsymbol{x}(j))\}
\end{aligned}
$$

Therefore, $p(\boldsymbol{x})$ is a feasible solution, and since α_is are positive and it is a maximization, it contradicts the optimality of $p_1(\boldsymbol{x})$ and $p_2(\boldsymbol{x})$. As a result, the optimal solution to (LP) is unique.

Now, we will demonstrate that the solution is independent of α. Again we use contradiction and assume $p_1(\boldsymbol{x})$ is optimal solution within $\boldsymbol{\alpha}^1$ and $p_2(\boldsymbol{x})$ is the optimal solution within $\boldsymbol{\alpha}^2$. Following the discussion above, the maximum of $p_1(\boldsymbol{x})$ and $p_2(\boldsymbol{x})$, $\forall \boldsymbol{x}$, is a feasible solution with a higher objective value, which contradicts the optimality of $p_1(\boldsymbol{x})$ and $p_2(\boldsymbol{x})$. Therefore, the optimal solution is unique and independent of α_is. ■

Proof Proposition 2: Let us use the superscript k to denote the last update in iteration k. We observe that the bundle prices and assignment from iteration k satisfy the following condition:

$$\sum_{x} y_i^{(k)}(\boldsymbol{x})(v_i(\boldsymbol{x}) - p^{(k)}(\boldsymbol{x})) \geq v_i(\boldsymbol{x}^{'}) - p^{(k)}(\boldsymbol{x}^{'}) \quad \forall i \in I, \boldsymbol{x}^{'} \in \{0,1\}^n \tag{C.1}$$

Let $\boldsymbol{X}^{(k)} := \{\boldsymbol{x} : \exists i, \boldsymbol{x}^{(k)}(i) = \boldsymbol{x}\}$ indicate the set of bundles that have been assigned to at least one of the consumers after the assortment step in iteration k. The pricing step in iteration $k+1$ solves the following optimization problem:

$$\begin{aligned}
&\max_{\tilde{p}(\boldsymbol{x})} \sum_{i \in \hat{I}^{(k)}} \alpha_i \tilde{p}(\boldsymbol{x}^{(k)}(i)) \\
&\quad s.t. v_i(\boldsymbol{x}^{(k)}(i)) - \tilde{p}(\boldsymbol{x}^{(k)}(i)) \geq v_i(\boldsymbol{x}^{(k)}(j)) - \tilde{p}(\boldsymbol{x}^{(k)}(j)) \quad \forall i, j \in \hat{I}^{(k)} \\
&\quad\quad v_i(\boldsymbol{x}^{(k)}(i)) \geq \tilde{p}(\boldsymbol{x}^{(k)}(i)) \quad \forall i \in \hat{I}^{(k)} \\
&\quad\quad \tilde{p}(\boldsymbol{x}^{(k)}(i)) \geq 0 \quad \forall i \in \hat{I}^{(k)}
\end{aligned}$$

Assume that there exists an optimal solution to the above problem such that there is $\boldsymbol{x}$ such that $p^{(k)}(\boldsymbol{x})) > p^{(k+1)}(\boldsymbol{x}))$. Inequalities (C.1) indicate that for $\boldsymbol{x} \in \boldsymbol{X}^{(k)}$, $p^{(k)}(\boldsymbol{x})$ is a feasible solution. Therefore, as demonstrated in the proof of Proposition 1, $\max\{p^{(k)}(\boldsymbol{x}), p^{(k+1)}(\boldsymbol{x})\}$ is also a feasible solution. Since the objective function is maximization and α_is are positive, it contradicts the optimality of $p^{(k+1)}(\boldsymbol{x})$. Thus, for $\boldsymbol{x} \in \boldsymbol{X}^{(k)}$, we have $p^{(k)}(\boldsymbol{x})) \leq p^{(k+1)}(\boldsymbol{x})$.

Furthermore, for $\boldsymbol{x} \notin \boldsymbol{X}^{(k)}$, we have:

$$p^{(k+1)}(\boldsymbol{x}) = \max\{v_i(\boldsymbol{x}) - (v_i(\boldsymbol{x}(i)) - p^{(k+1)}(\boldsymbol{x}(i)) : i \in \hat{I}^{(k)}\} \leq \max\{v_i(\boldsymbol{x}) - (v_i(\boldsymbol{x}(i)) - p^{(k)}(\boldsymbol{x}(i)) : i \in \hat{I}^{(k)}\}$$

where the inequality follows from the first part of the proof.

■

Proof Proposition 3: We will prove at least one consumer types has a zero surplus by using contradictions. Consider all consumers have a positive surplus and $p^*(\boldsymbol{x})$ is an optimal solution to (LP). Let $i_{\min} = \arg\min\{su_i : i \in \hat{I}\}$ indicate the consumer type in $\hat{I}$ with the minimum surplus. Consider there is another solution such that $p(\boldsymbol{x}) = p^*(\boldsymbol{x}) + su_{i_{\min}}$. As can be seen, $p(x)$ satisfies the first constraint (IC), since only constant $sui_{\min}$ is subtracted from both sides of the inequality. According to the definition of su_i, the following is implied:

$$\begin{aligned}
v_i(\boldsymbol{x}(i)) - p^*(\boldsymbol{x}(i)) \geq & su_i \qquad \forall i \in \hat{I} \\
& \Rightarrow v_i(\boldsymbol{x}(i)) - p^*(\boldsymbol{x}(i)) \geq su_{i_{\min}} \\
& \Rightarrow v_i(\boldsymbol{x}(i)) \geq p^*(\boldsymbol{x}(i)) + su_{i_{\min}}
\end{aligned}$$

Thus, $p(x)$ also satisfies the second constraint (IR) and is a feasible solution. α_is are positive, so $\sum_{i\in\hat{I}} \alpha_i p^*(\boldsymbol{x}(i)) \leq \sum_{i\in\hat{I}} \alpha_i p(\boldsymbol{x}(i))$, which contradicts the optimality of $p^*(\boldsymbol{x})$s. Therefore, at least one of the consumers has a zero surplus.

The IR constraint ensures that consumers have a non-negative surplus. Therefore, if the kth iteration of the PAH algorithm extracts all consumer surplus by repeating the algorithm, it cannot increase the prices. Furthermore, based on the Proposition 2 bundle prices are non-decreasing in the PAH algorithm, so repeating the algorithm cannot change the prices. Thus, changing the assignment cannot improve the seller's profit, and the algorithm will terminate. ■

Proof Proposition 4: As a starting point, at least one bundle must be assigned to one consumer type, such as bundle $\boldsymbol{x}$ to type i. This implies that $p(\boldsymbol{x}) \leq v_i(\boldsymbol{x})$. Therefore:

$R^0 \geq \frac{1}{|I|} \min_{\boldsymbol{x},i}\{v_i(\boldsymbol{x})\}$

Additionally, IR constraints indicate that the price of the bundle that consumers purchase is less than $v_{\max}$. Accordingly:

$R_P \leq v_{\max} \Rightarrow \frac{R_P}{R^0} \leq \frac{|I|v_{\max}}{v_{\min}}$

We consider two products in which one has a value of zero for all consumers, while the other only has a value of $v_{\min}$ for consumer type i and zero for the rest. The grand bundle of two products has a value of $v_{\max}$ for all consumer types. Assigning product 2 to consumer type i is feasible point, however after solving one iteration of Steps 1 and 2, the grand bundle is assigned to all consumers at the price of $v_{\max}$. Hence, $\frac{R_P}{R^0} = \frac{|I|v_{\max}}{v_{\min}}$.

As a result, the bound is tight. ■

Proof Lemma 1: Let us consider a scenario with two consumer types with equal probability and two products. Consider bundles product and cost be as follow:

$$
\begin{aligned}
&c(1,0) = \epsilon,\ c(0,1) = c - \epsilon,\ c(1,1) = c \\
&v_1(1,0) = c,\ v_1(0,1) = \epsilon,\ v_1(1,1) = c + \epsilon \\
&v_2(1,0) = 0,\ v_2(0,1) = c + \epsilon,\ v_2(1,1) = c + \epsilon.
\end{aligned}
$$

The optimal solution of pure bundling sets the grand bundle price at $c + \epsilon$, resulting in an expected profit of 2ϵ. To evaluate the performance of the PAH algorithm in this context, the problem is first transformed into one with zero cost, denoted transformed reservation prices as $v^{'}(.)$. This transformation yields the following reservation prices:

$$
\begin{aligned}
&v_1^{'}(1,0) = c - \epsilon,\ v_1^{'}(0,1) = 2\epsilon - c,\ v_1^{'}(1,1) = \epsilon \\
&v_2^{'}(1,0) = -\epsilon,\ \ v_2^{'}(0,1) = 2\epsilon,\ \ v_1^{'}(1,1) = \epsilon
\end{aligned}
$$

In the first iteration of the PAH algorithm, the prices for the products are set as $p(1,0) = c - \epsilon$,

$p(0,1) = 2\epsilon$, and $p(1,1) = \epsilon$. It is important to note that the prices given by the PAH algorithm are after subtracting the cost, and the exact price must be determined by adding the cost, i.e., the algorithm sets the price of both grand bundle and product 2 at $c + \epsilon$.

The PAH algorithm assigns bundle $(1,0)$ with revenue $c - \epsilon$ to consumer type 1, and bundle $(0,1)$ with revenue 2ϵ to consumer type 2, resulting in a total profit of $c + \epsilon$. It is worth noting that the PAH algorithm terminates after a single iteration, as it pushes all consumer surpluses to zero. However, the expected profit from pure bundling is 2ϵ. As a result, the expected profit from the PAH algorithm relative to that of pure bundling is $\frac{c+\epsilon}{2\epsilon}$.

As ϵ approaches zero, the relative expected profit from the PAH algorithm increases without bound, as shown by the expression $\frac{c+\epsilon}{2\epsilon} \to \infty$. Therefore, even when considering additive buyer and additive cost, the PAH algorithm has the potential to improve expected profits arbitrarily. ■

Proof Proposition 5: We will begin by comparing the upper bound of mixed bundling, which serves as an upper bound for the PAH algorithm, with the optimal solution of pure bundling. We will then provide an instance of the problem in which the PAH algorithm can achieve the upper bound.

Assume that the consumers' reservation prices for the grand bundle are ordered, such that $v_{(1)}(e)$ refers to the highest reservation price for the grand bundle, where e is a vector of ones. Without loss of generality, let us assume that the optimal solution of pure bundling is to offer a grand bundle at a price such that $1 \leq k \leq |I|$ consumer types will purchase the grand bundle, and the rest will leave without purchasing. It is easy to see that the optimal price would be $v_{(k)}$. Its optimality implies that:

$$kv_{(k)} \geq lv_{(l)} \qquad \forall l \in I \tag{C.2}$$

A clairvoyant with information about consumer types can offer a personalized price for each consumer, extracting all consumer surplus. Let R_C denote the expected profits of a clairvoyant with the ability to offer personalized pricing. Then:

$$R_C = \sum_{l \in I} v_{(l)} \leq \sum_{l \in I} \frac{kv_{(k)}}{l}$$

The inequality is due to (C.2). Let $R_{P,u}$ represent the seller profit produced by PAH algorithm given optimal solution of pure bundling as an initial point. We have:

$$\frac{R_{P,u}}{R_u} \leq \frac{R_C}{R_u} \leq \frac{\sum\limits_{l \in I} \frac{kv_{(k)}}{l}}{kv_{(k)}} = \sum_{l \in I} \frac{1}{l} = \ln|I| + \gamma + \frac{1}{2|I|} - \varepsilon_{|I|} < 1 + \ln(|I|) \tag{C.3}$$

Where $\gamma \approx 0.5772$ is the Euler–Mascheroni constant and $0 \leq \varepsilon_{|I|} \leq \frac{1}{8|I|^2}$. The last inequality is due to

$$|I| \to \infty \quad \Rightarrow \frac{1}{2|I|}, \varepsilon_{|I|} \to 0 \quad \Rightarrow \ln|I| + \gamma + \frac{1}{2|I|} - \varepsilon_{|I|} \to \ln|I| + \gamma$$

We will now demonstrate that the bound $\frac{R_{P,u}}{R_u} \leq \sum\limits_{l \in I} \frac{1}{l}$ is achievable. To do so, we will provide an example of the problem in which the PAH algorithm can achieve the expected profits of a clairvoyant starting from the optimal solution of pure bundling.

Let $e_i \in \{0,1\}^n$ indicate a vector with one in the ith position and 0s elsewhere, and e denote vector of ones. Let us consider the following scenario: The number of consumers is equal to the number of products, that is, $n = |I|$, and the reservation prices of the consumers are as follows:

$$v_i(\boldsymbol{x}) = \begin{cases} \frac{v-\mathbb{1}_{i\neq n}\epsilon}{n-i+1} & \boldsymbol{x} = e - e_i \\ 0 & \text{otherwise} \end{cases} \qquad \forall i \in I$$

The pure bundling strategy involves offering a grand bundle at a price of v, with a corresponding profit of v. Considering this as an initial point for PAH algorithm. In the first iteration, the price of the bundle $e - e_n$ and the grand bundle are assigned at v, while the prices of the remaining bundles are set to zero. In the assortment step of the first iteration, the bundle $e - e_i$ is assigned to consumer $i \neq n$, while the grand bundle is assigned to consumer n. In the second iteration, the price of the bundle $e - e_i$ is increased to $\frac{v-\epsilon}{n-i+1}$. The algorithm terminates after this step because all consumer surplus has been pushed to zero. According to the proposition, if the PAH algorithm has extracted all consumer surplus, then no further improvement is possible, and the algorithm will terminate. Therefore, we have:

$$\lim_{\epsilon\to 0} \frac{R_{P,u}}{R_u} = \lim_{\epsilon\to 0} \frac{\sum_{i\in I} \frac{v-\mathbb{1}_{i\neq n}\epsilon}{n-i+1}}{v} = \lim_{\epsilon\to 0} \frac{\sum_{i\in I} (v - \mathbb{1}_{i\neq 1}\epsilon)\frac{1}{i}}{v} = \sum_{i\in I} \frac{1}{i}$$

This example demonstrates that the PAH algorithm can achieve the expected profits of a clairvoyant starting from the optimal solution of pure bundling, and thus, the bound $\frac{R_{P,u}}{R_u} \leq \sum_{l\in I} \frac{1}{l}$ is achievable. ■

Proof Corollary 1: As pure bundling is a special case of BSP, so $R_b \geq R_u$. Therefore,

$$\frac{R_{P,b}}{R_b} \leq \frac{R_C}{R_b} \leq \frac{R_C}{R_u} \leq \sum_{i\in I} \frac{1}{i}$$

The last inequality is due to the Inequality(C.3).

Next, we will demonstrate that the example given in the proof of Proposition 5 yields the same solution for BSP as pure bundling, allowing us to attain the bound $\frac{R_{P,b}}{R_b} \leq \sum_{l\in I} \frac{1}{l}$. The consumer reservation prices for bundle sizes are as follows:

$$v_{n-i}(k) = \frac{v-\epsilon}{i+1} \quad n-1 \leq k, 1 \leq i \leq n-1$$
$$v_n(k) = v \qquad n-1 \leq k$$
$$v_i(k) = 0 \qquad \forall i \in I, k \leq n-2$$

It is easy to see that the BSP assign grand bundle at price v to consumer type n with a profit v. ■

Proof Lemma 2: The third constraint in (PA_k) implies that if there exists a $\boldsymbol{x} \in \boldsymbol{X}k$ such that $y_i(\boldsymbol{x}) = 1$, then:

$$su(i) \leq v_i(\boldsymbol{x}) - p(\boldsymbol{x}) \Rightarrow p(\boldsymbol{x}) \leq v_i(\boldsymbol{x}) - su(i) \leq \max \boldsymbol{x} \in \boldsymbol{X}_k v_i(\boldsymbol{x}) - su(i)$$

The non-negativity constraint on prices implies that $pp_i \geq 0$. Assume that there exists a $pp_i = 0$, which implies that there is a bundle with price zero that is assigned to one of the consumers. We will show that in the optimal solution of the problem, all bundle prices are strictly positive.

Consider $p^*(\boldsymbol{x})$ to be an optimal solution of problem (PA_k) such that $p^*(\boldsymbol{x}^{'}) = 0$. Now consider another solution where for $\boldsymbol{x} \neq \boldsymbol{x}^{'}$, $p(\boldsymbol{x}) = p(\boldsymbol{x})$, and $p(\boldsymbol{x}^{'} = \epsilon) > 0$. It is easy to see that this solution does not change the assignment for consumers who purchase products except for those who purchase $\boldsymbol{x}^{'}$, and the expected profit from these consumers is zero, so at least has a same expected profit as optimal solution. Moreover, as equation (1) illustrates, increasing the price of some bundles cannot decrease the price of other bundles, so the profit from the new solution is an upper bound for the profit from $p^*(\boldsymbol{x})$, which contradicts its optimality. Therefore, in the optimal solution, no bundle has a zero price. ■

D Shapley Value as an Starting point

We measure the marginal contribution of each product for each consumer type by computing $s_n^i(\boldsymbol{x}) = v_i(\boldsymbol{x} + \boldsymbol{e}_n) - v_i(\boldsymbol{x})$. The Shapley value of each product for each consumer type is then computed as follows:

$$s_n^i := \sum_{\boldsymbol{x}} \frac{(\boldsymbol{e}^T \boldsymbol{x})!\,(n - \boldsymbol{e}^T \boldsymbol{x} - 1)!}{n} s_n^i(\boldsymbol{x}), \qquad \forall n \in N$$

Calculating s_n^i involves numerous non-trivial computations due to the factorial complexity, which is non-polynomial time. However, Proposition D.1 simplifies the computation of the Shapley values. Let $\boldsymbol{a}_n$ denotes row n of matrix $\boldsymbol{A}$.

Proposition D.1. *Suppose that $v(\boldsymbol{x}) = \boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x}$, where $\boldsymbol{A}$ is a symmetric matrix then*
$s_n = \boldsymbol{e}_n^T \boldsymbol{A} \boldsymbol{e} = \boldsymbol{a}_n \boldsymbol{e} = \sum_j a_{nj}$.

Proof Proposition D.1

Here abusing notation $N^{'} := \{0,1\}^n$, and $\boldsymbol{x}_n$ indicates the nth position of the vector $\boldsymbol{x}$:
$v(\boldsymbol{x} + \boldsymbol{e}_n) - v(\boldsymbol{x}) = (\boldsymbol{x}^{'} + \boldsymbol{e}_n{'})\boldsymbol{A}(\boldsymbol{x} + \boldsymbol{e}_n) - \boldsymbol{x}^{'}\boldsymbol{A}\boldsymbol{x} = (\boldsymbol{x}^{'}\boldsymbol{A} + \boldsymbol{e}_n^{'}\boldsymbol{A})(\boldsymbol{x} + \boldsymbol{e}_n) - \boldsymbol{x}^{'}\boldsymbol{A}\boldsymbol{x} = \boldsymbol{x}^{'}\boldsymbol{A}\boldsymbol{x} + \boldsymbol{e}_n^{'}\boldsymbol{A}\boldsymbol{x} + \boldsymbol{x}^{'}\boldsymbol{A}\boldsymbol{e}_n + \boldsymbol{e}_n^{'}\boldsymbol{A}\boldsymbol{e}_n - \boldsymbol{x}^{'}\boldsymbol{A}\boldsymbol{x} = a_{nn} + 2\boldsymbol{e}_n^{'}\boldsymbol{A}\boldsymbol{x}$
$\Rightarrow s_n = \sum_{\boldsymbol{x} \in N^{'}:\boldsymbol{x}_n=0} \frac{(\boldsymbol{e}^{'}\boldsymbol{x})!(n-\boldsymbol{e}^{'}\boldsymbol{x}-1)!}{n!}[a_{nn} + 2\boldsymbol{e}_n^{'}\boldsymbol{A}\boldsymbol{x}] = a_{nn}\sum_{\boldsymbol{x} \in N^{'}:\boldsymbol{x}_n=0} \frac{(\boldsymbol{e}^{'}\boldsymbol{x})!(n-\boldsymbol{e}^{'}\boldsymbol{x}-1)!}{n!} + 2\sum_{\boldsymbol{x} \in N^{'}:\boldsymbol{x}_n=0} \frac{(\boldsymbol{e}^{'}\boldsymbol{x})!(n-\boldsymbol{e}^{'}\boldsymbol{x}-1)!}{n!}\boldsymbol{a}_n\boldsymbol{x}$

Where the first summation is simplified as follows:

$\sum_{\boldsymbol{x}\in N':\boldsymbol{x}_n=0} \frac{(\boldsymbol{e}'\boldsymbol{x})!(n-\boldsymbol{e}'\boldsymbol{x}-1)!}{n!} = \sum_{\boldsymbol{x}\in N':\boldsymbol{x}_n=0} \frac{(|\boldsymbol{x}|)!(n-|\boldsymbol{x}|-1)!}{n!}$

$= \sum_{\boldsymbol{x}\in N':\boldsymbol{x}_n=0} \frac{1}{n} \binom{n-1}{|\boldsymbol{x}|}^{-1} = \frac{1}{n} \sum_{|\boldsymbol{x}|=0}^{n-1} \binom{n-1}{|\boldsymbol{x}|}^{-1} \times \binom{n-1}{|\boldsymbol{x}|} = \frac{1}{n} \sum_{|\boldsymbol{x}|=0}^{n-1} 1 = \frac{n}{n} = 1$

$\Rightarrow s_n = a_{nn} + 2\boldsymbol{a}_n [\sum_{\boldsymbol{x}\in N':\boldsymbol{x}_n=0} \frac{(|\boldsymbol{x}|)!(n-|\boldsymbol{x}|-1)!}{n!} \boldsymbol{x}]$

$\sum_{\boldsymbol{x}\in N':\boldsymbol{x}_i=0} \frac{(|\boldsymbol{x}|)!(n-|\boldsymbol{x}|-1)!}{n!} \boldsymbol{x} = \frac{1}{n} [\sum_{j\neq i} \binom{n-1}{1}^{-1} \boldsymbol{e}_j + \sum_{j\neq n} \sum_{k\neq j,n} \binom{n-1}{2}^{-1} (\boldsymbol{e}_j + \boldsymbol{e}_k) + \ldots + \binom{n-1}{n-1} \boldsymbol{e}_n] = \frac{1}{n} \sum_{j\neq n} \{ \binom{n-1}{1}^{-1} + \binom{n-1}{2}^{-1} \binom{n-2}{1} + \ldots + \binom{n-1}{n-1}^{-1} \binom{n-2}{0} \} \boldsymbol{e}_j = \frac{1}{n} \sum_{j\neq n} \{ \frac{1}{n-1} + \frac{2}{n-1} + \ldots + \frac{n-1}{n-1} \} \boldsymbol{e}_j = \frac{1+2+\ldots+n-1}{n(n-1)} \sum_{j\neq n} \boldsymbol{e}_j = \frac{1}{2} \sum_{j\neq n} \boldsymbol{e}_j = \frac{\boldsymbol{e}_n}{2}$

$\Rightarrow s_n = a_{nn} + \boldsymbol{a}_n \boldsymbol{e}_n = \boldsymbol{a}_n \boldsymbol{e}$ ■

Proposition D.1 is similar to the results obtained by Colini-Baldeschi et al. (2018) for variance matrix, which is symmetric positive semi-definite matrix. Wang (2002) and Owen and Prieur (2017) found similar results for variance matrices of multivariate normal distributions and variance matrices of linear regressions with multivariate variables. However, we generalize all previous results, showing that it is correct for any symmetric matrix, regardless of its element.

The creation of value by each product can be viewed as a cooperative effect of the products offered. The pricing challenge involves determining the optimal price for a product or service that maximizes profit or revenue while taking into account market demand and customer reservation price. To tackle these challenges, we adopt a game-theoretic perspective, where each product contributor is modeled as a player in a cooperative game. Therefore, the price of products is determined by a weighted average of their Shapley values

$$p(\boldsymbol{e}_n) = \max\{\sum_i \alpha_i s_n^i,\ c(\boldsymbol{e}_n)\} = \max\{\sum_i \alpha_i \sum_j a_{nj},\ c(\boldsymbol{e}_n)\}$$

Where consumers are free to choose any bundle $\boldsymbol{x}$ at price $\boldsymbol{x}^T\boldsymbol{p}$.

As achieving an optimal solution for component pricing, even with symmetric matrices, can be difficult, we explore the effectiveness of the Shapley value as a heuristic for component pricing. Specifically, we compute the Shapley value and employ the scaling strategy proposed by Gallego and Berbeglia (2021) as a heuristic for component pricing. This strategy aims to identify the optimal price along a positive vector, which reduces the number of decision variables to just one continuous variable related to scaling rather than the n continuous variables related to price. A numerical study involving up to five products and around two million data points indicates that this approach can yield approximately 80% of the component pricing profit for a firm. However, further analytical investigation is required to confirm the profitability of the Shapley value, which is beyond the scope of this study.